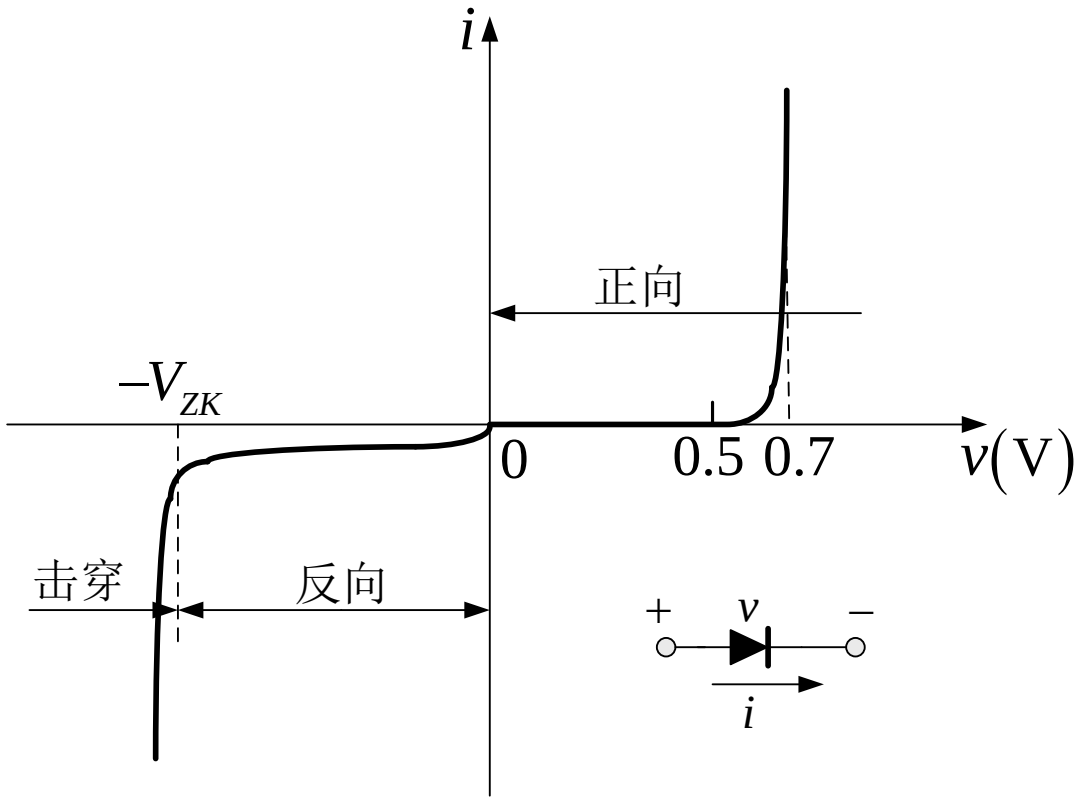
期末复习

二 极管及其应用电路

1、二极管伏安特性及其常用等效电路



二极管模型

$$i = I_S\left(e^{\frac{v}{nV_T}} - 1\right) \quad \Longrightarrow$$

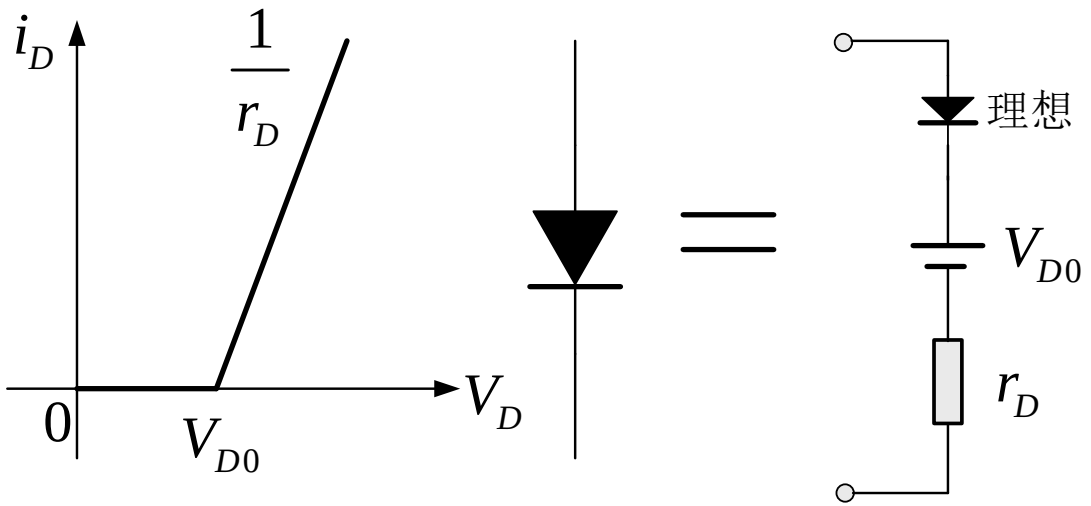
$$V_2 - V_1 = nV_T \ln\frac{I_2}{I_1} = \frac{1}{\log e} nV_T \log\frac{I_2}{I_1} \approx 2.3nV_T \log\frac{I_2}{I_1}$$

室温下，$V_T = kT/q \approx 25\ \text{mV}$（或 26 mV）

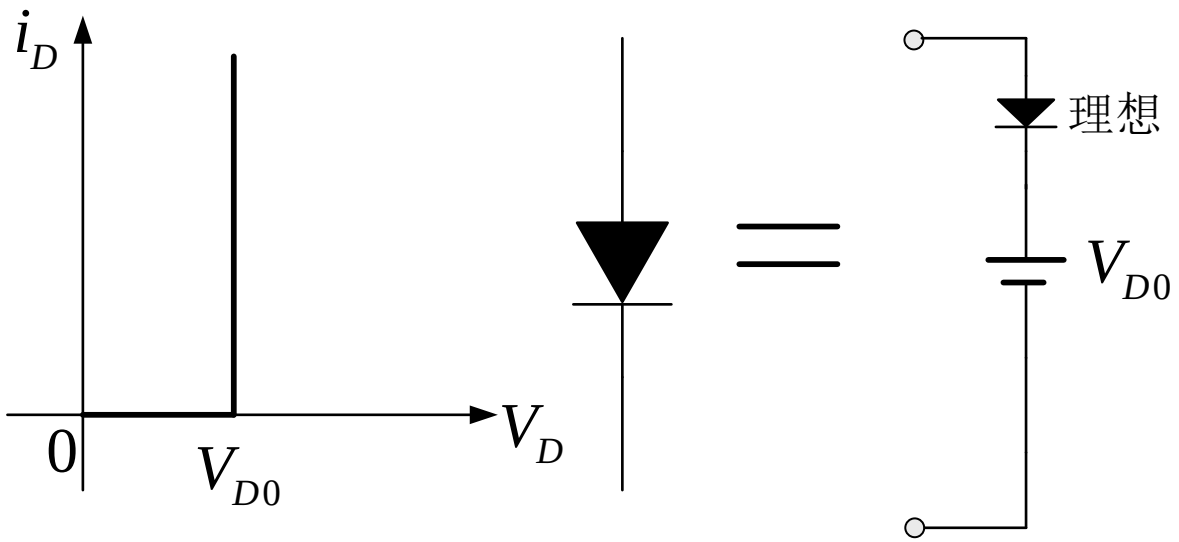
***n*=1**（集成二极管）

***n*=2**（分立二极管）

简化二极管模型

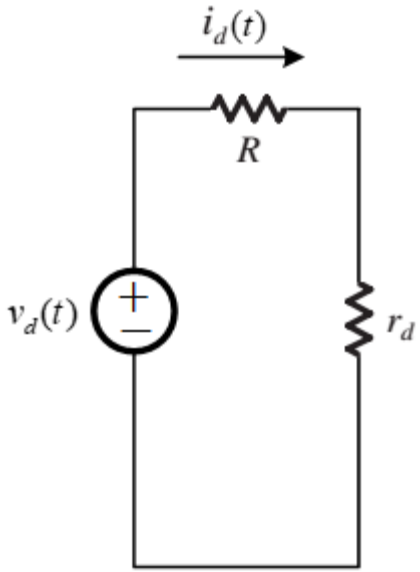


恒压降模型

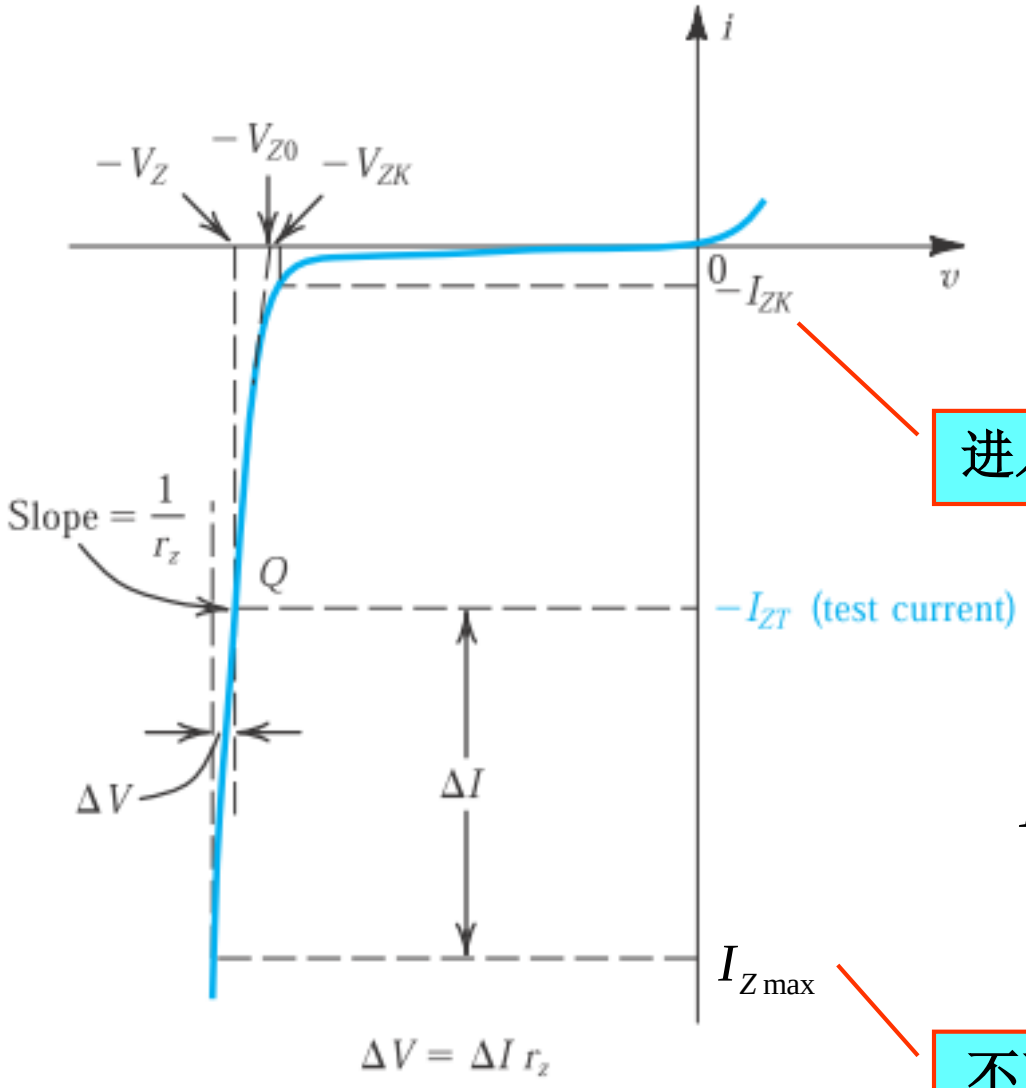


小信号模型

$$r_d = \frac{v_d}{i_d} = \frac{nV_T}{I_D}$$



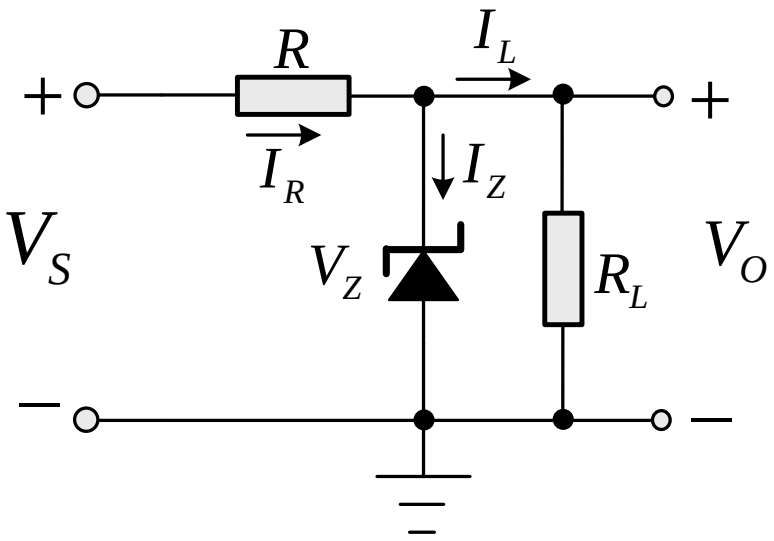
2、稳压二极管伏安特性及其稳压电路



进入稳压区的最小电流

$$I_{ZK} < I_Z < I_{Z\max}$$

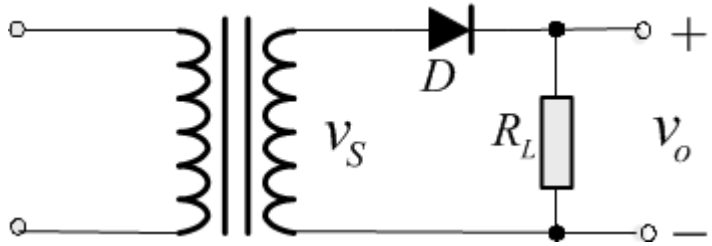
不至于损坏的最大电流



$$I_{Zk} < \frac{V_I - V_Z}{R} - I_L < I_{\max}$$

3、整流电路

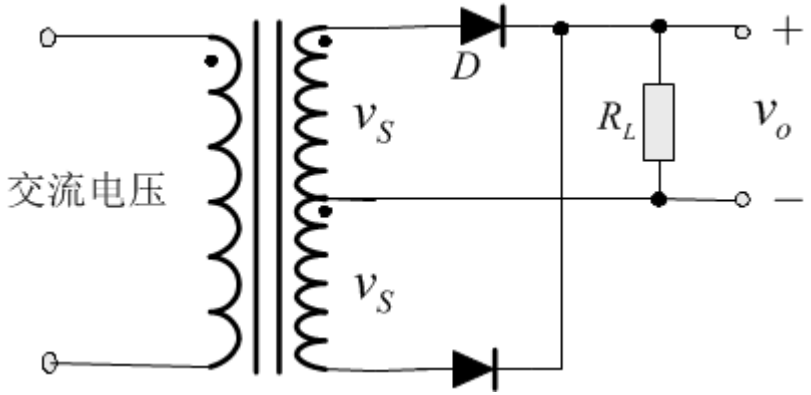
半波整流器



$$PIV = V_{\mathrm{om}}$$

$$V_O = \frac{\int_0^{\pi} \sqrt{2} V_{orms} \sin\phi \mathrm{d}\phi}{2\pi} \approx 0.45 V_{orms}$$

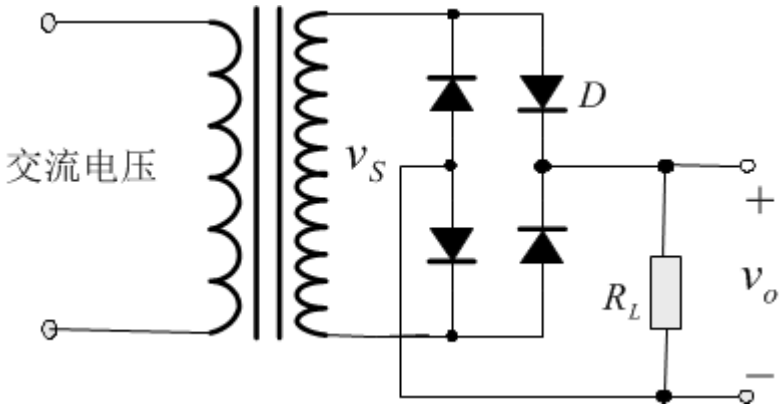
全波整流器



$$PIV = 2V_{\mathrm{om}} - V_D \approx 2V_{\mathrm{om}}$$

$$V_O = \frac{\int_0^{\pi} \sqrt{2} V_{orms} \sin\phi \mathrm{d}\phi}{\pi} \approx 0.9 V_{orms}$$

桥式整流器

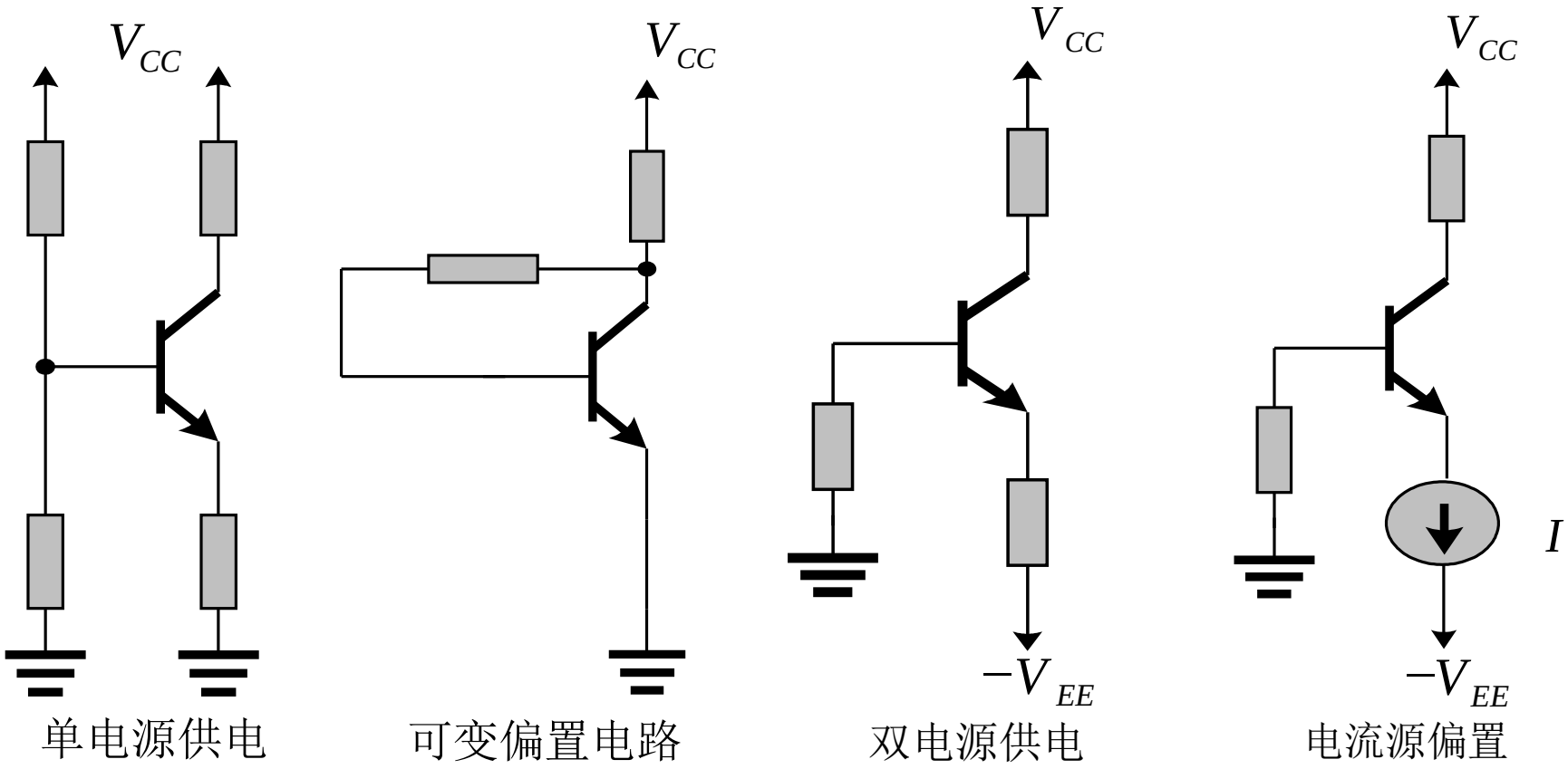


$$PIV = V_{\text{om}} - V_D \approx V_{\text{om}}$$

$$V_O = \frac{\int_0^{\pi} \sqrt{2} V_{orms} \sin\phi \, \mathrm{d}\phi}{\pi} \approx 0.9 V_{orms}$$

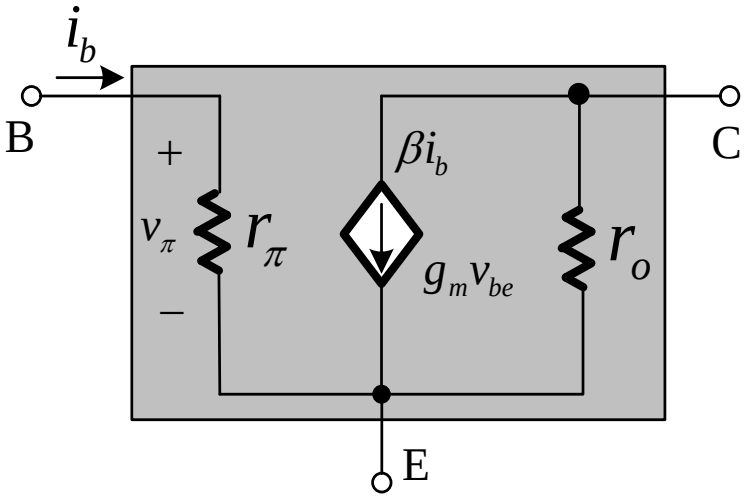
BJT及其放大电路

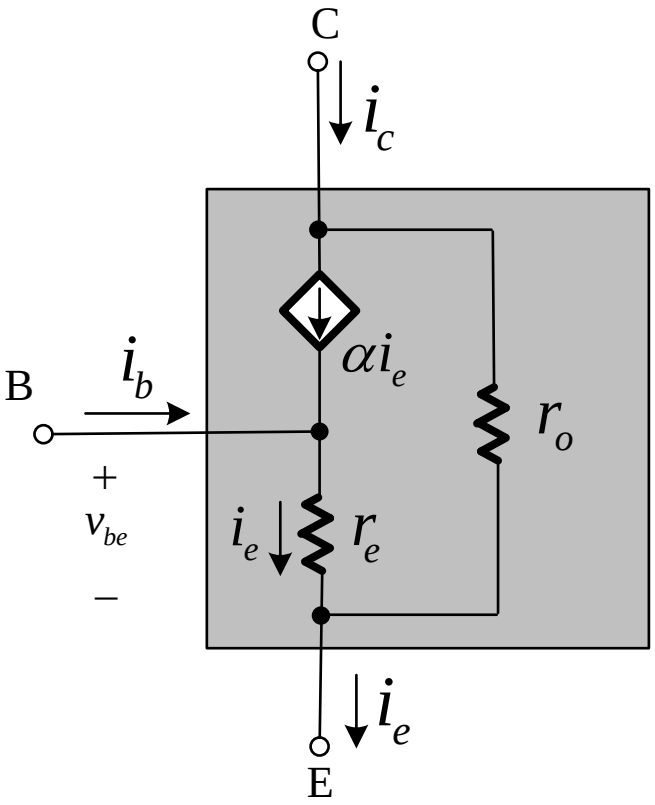
1、直流偏置电路及其分析



单电源供电　　可变偏置电路　　双电源供电　　电流源偏置

2、 π型和T型等效电路





3、 三种接法放大电路的分析计算

$$g_m = \frac{i_c}{v_{be}} = \frac{I_C}{V_T}$$

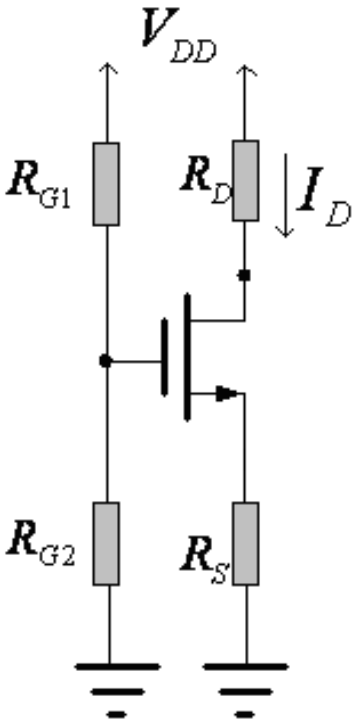
$$r_\pi = \frac{v_{be}}{i_b} = \frac{\beta}{g_m} = \frac{V_T}{I_B}$$

$$r_e = \frac{v_{be}}{i_e} = \frac{V_T}{I_E} = \frac{\alpha}{g_m} \approx \frac{1}{g_m}$$

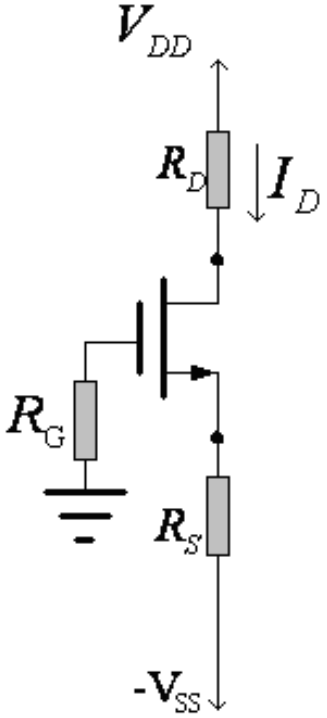
接法	共射	共集	共基
A_v	大 (<0)	~ 1	大
A_i	β	1＋β	α
R_i	中	大	小
R_o	大	小	大
频带	窄	中	宽

MOS管及其放大电路

1、直流偏置电路及其分析

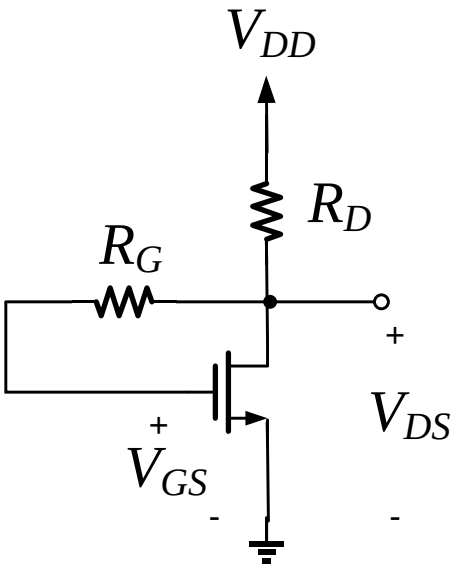


单电源

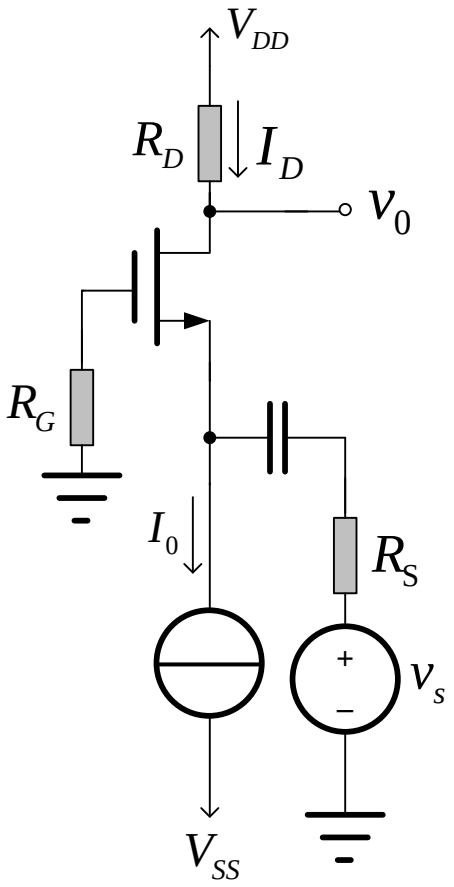


双电源

自给偏置电路



栅源间接反馈电阻的偏置电路



恒流源偏置电路

漏极电流（以NMOS为例）

变阻区　$v_{DS} < v_{GS} - V_{tn}$　或　$v_{GD} > V_{tn}$

$$i_D = k_n' \frac{W}{L}\left[\left(v_{GS} - V_{tn}\right)v_{DS} - \frac{1}{2}v_{DS}^2\right]$$

饱和区　$v_{DS} \geq v_{GS} - V_{tn}$　或　$v_{GD} \leq V_{tn}$

$$i_D = \frac{1}{2}k_n' \frac{W}{L}\left(v_{GS} - V_{tn}\right)^2$$

对于增强型PMOS管， $V_{tp}<0$　　$v_{GS}<0$　　$v_{DS}<0$

变阻区

$$v_{DS}>v_{GS}-V_{tp} \quad 或 \quad v_{GD}<V_{tp}$$

$$|v_{DS}|<|v_{GS}|-|V_{tp}| \quad 或 \quad |v_{GD}|>|V_{tp}|$$

$$i_D=k_p'\frac{W}{L}\left[\left(v_{GS}-V_{tp}\right)v_{DS}-\frac{1}{2}v_{DS}^2\right]$$

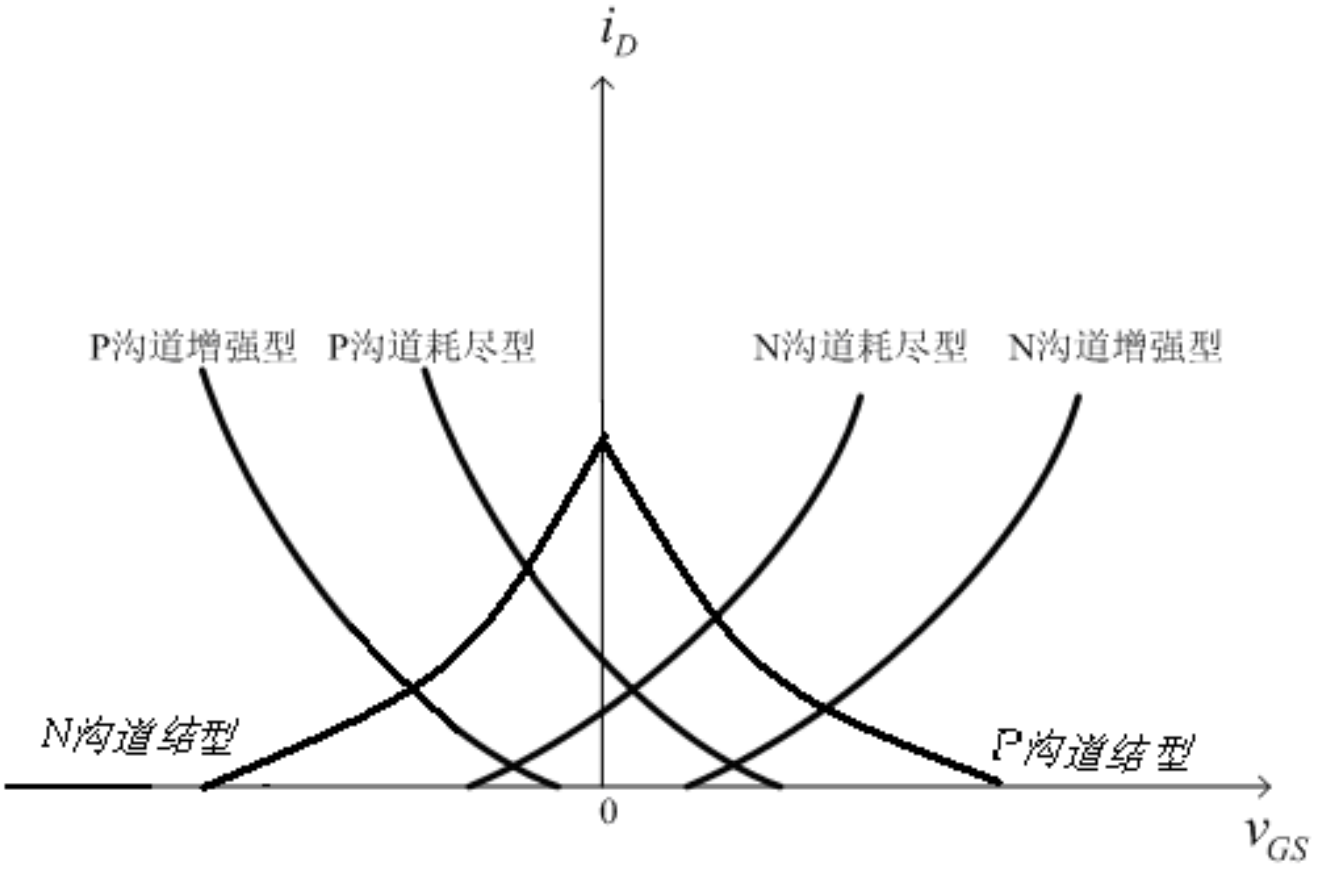
饱和区

$$v_{DS}\le v_{GS}-V_{tp} \quad 或 \quad v_{GD}\ge V_{tp}$$

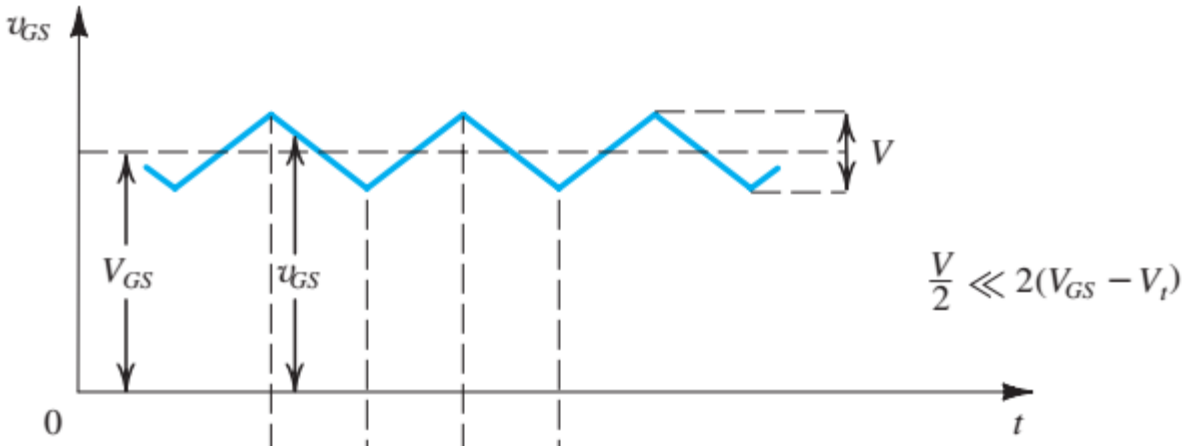
$$|v_{DS}|\ge|v_{GS}|-|V_{tp}| \quad 或 \quad |v_{GD}|\le|V_{tp}|$$

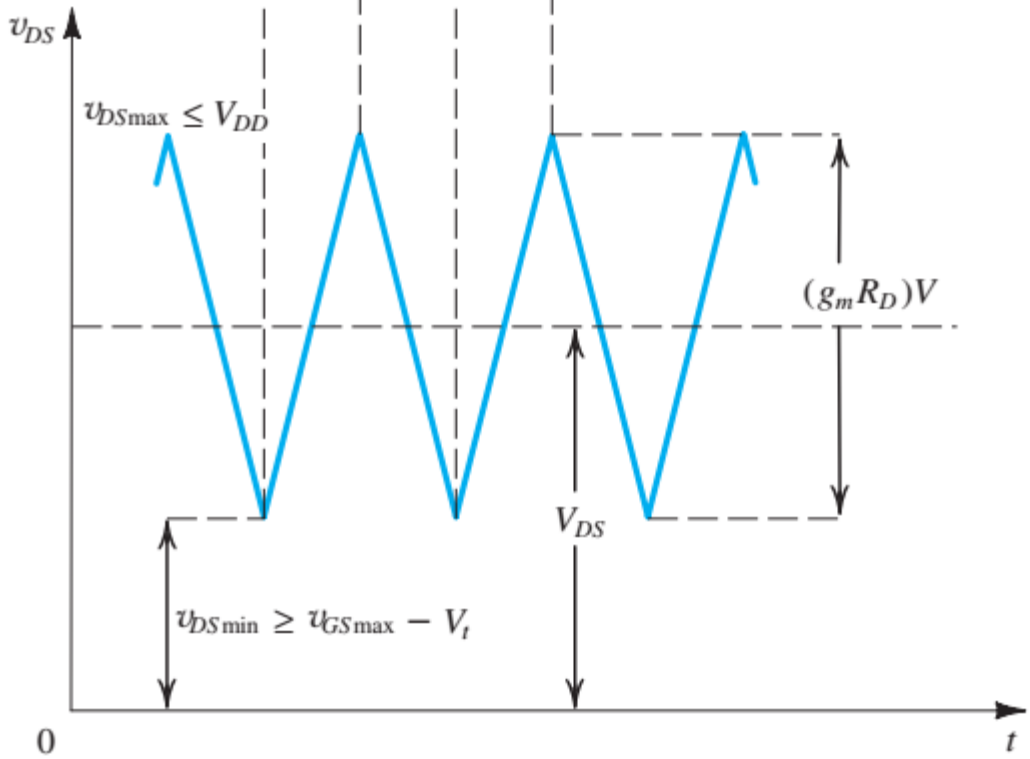
$$i_D=\frac{1}{2}k_p'\frac{W}{L}\left(v_{GS}-V_{tp}\right)^2$$

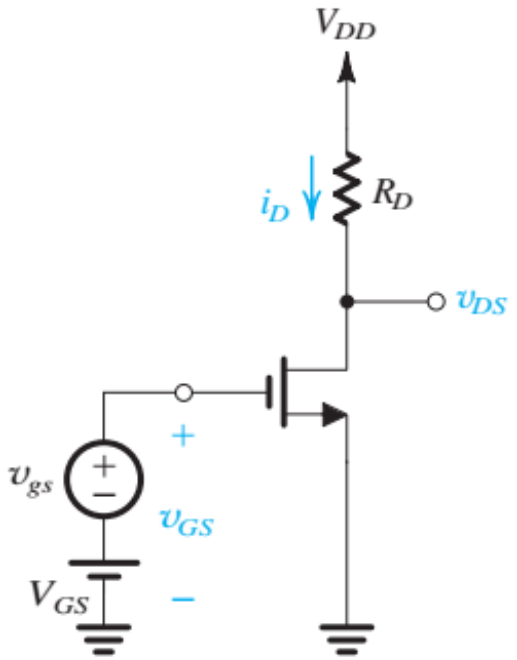
6种场效应管的转移特性曲线



输出电压 v_{DS} 的范围是否有限制？

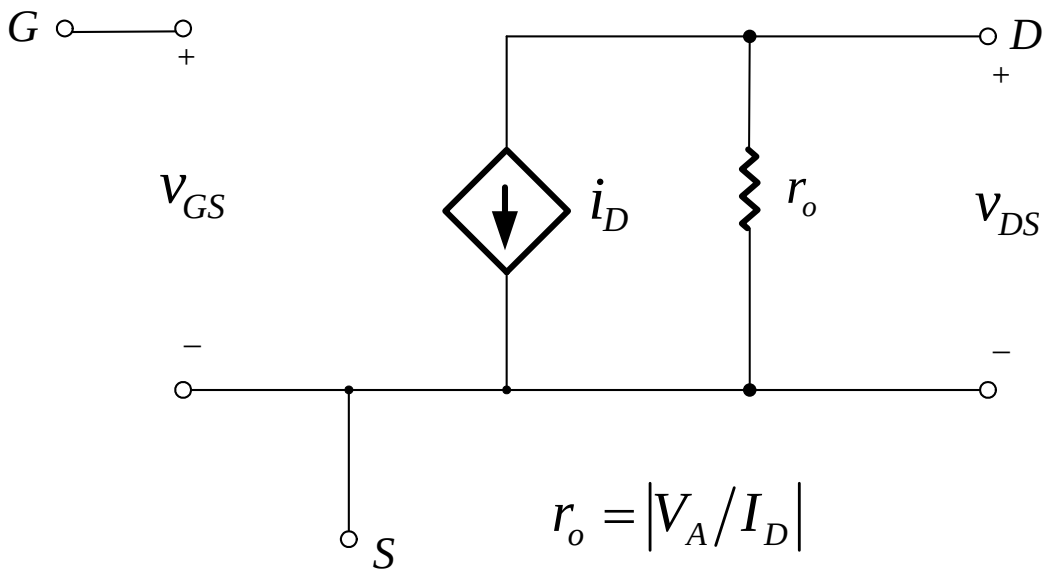






2、π型和T型等效电路

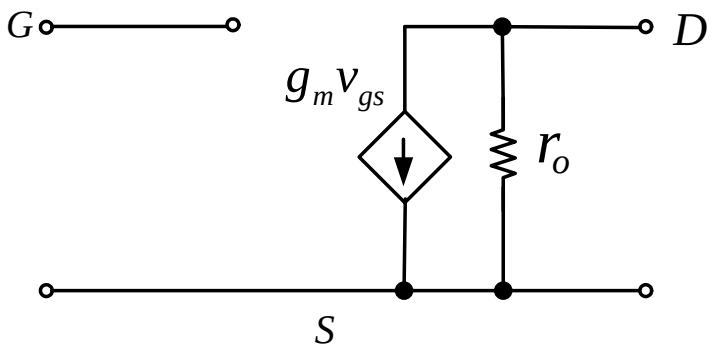
一、大信号模型

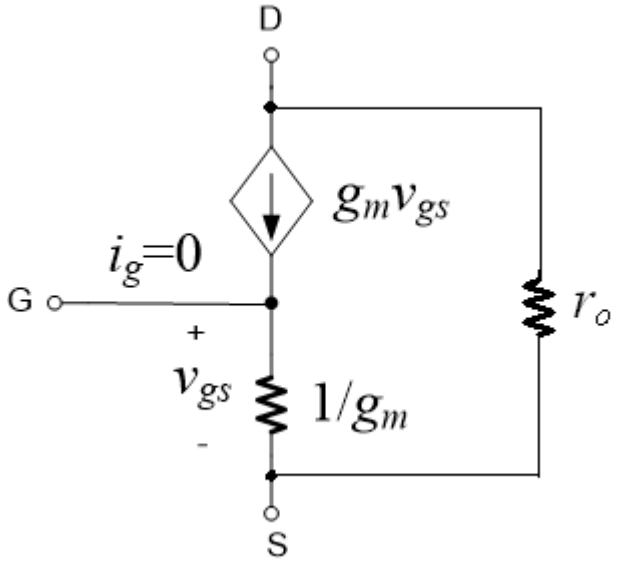


$$r_o = \left| V_A / I_D \right|$$

$$i_D = \frac{1}{2} k_n' \frac{W}{L} (v_{GS} - V_t)^2$$

二、小信号模型





3、三种接法放大电路的分析计算

$$g_m = \frac{i_d}{v_{gs}} = k_n' \frac{W}{L}(V_{GS} - V_t) \qquad g_m = \frac{I_D}{(V_{GS} - V_t)/2}$$

接法	共源	共漏	共栅
A_v	适中	~1	适中
A_i	/	/	~1
R_i	大	大	小
R_o	大	小	大
频带	窄	中	宽

运算放大器

1. 运放的理想化条件

(1) 差模增益趋于无穷大（$A \to \infty$）

(2) 输入阻抗趋于无穷大 （ $R_i \to \infty$）

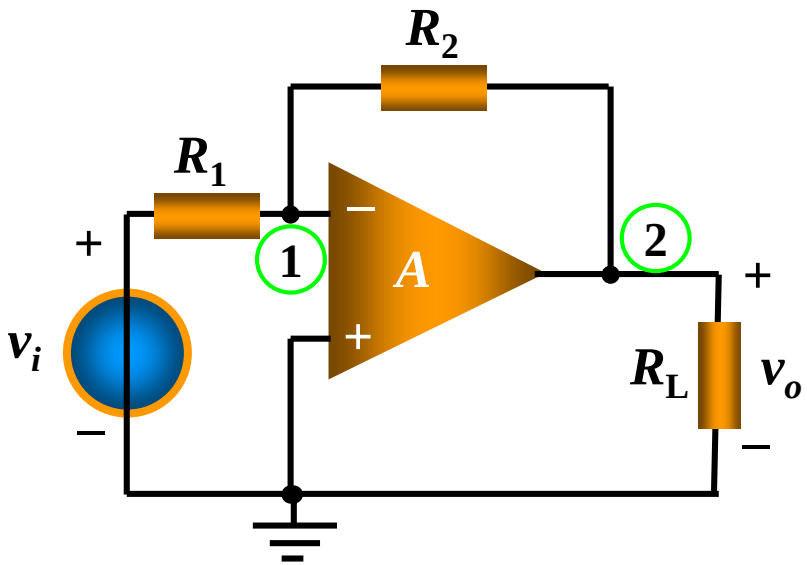
(3) 输出阻抗趋于零 （ $R_o \to 0$）

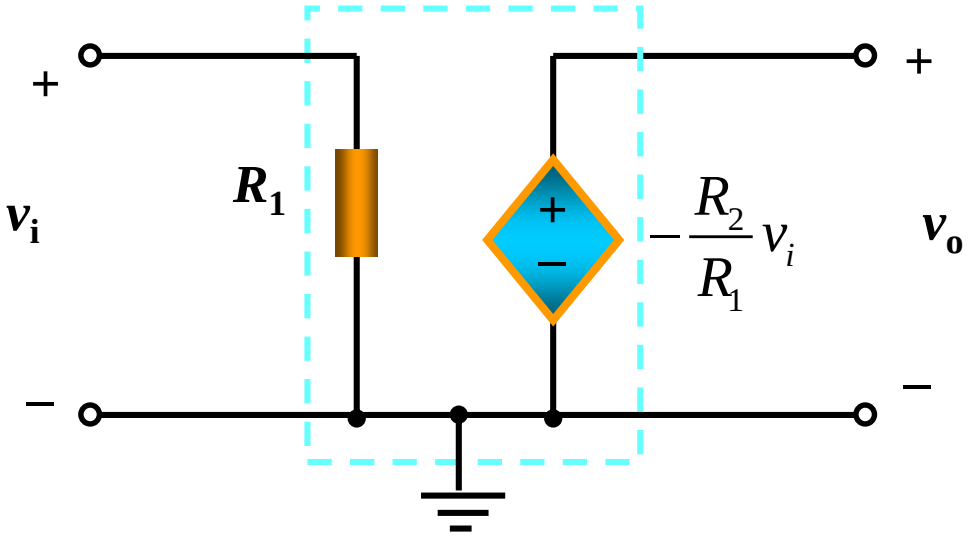
(4) 共模抑制比趋于无穷大（$CMRR \to \infty$）

(5) 带宽无穷大

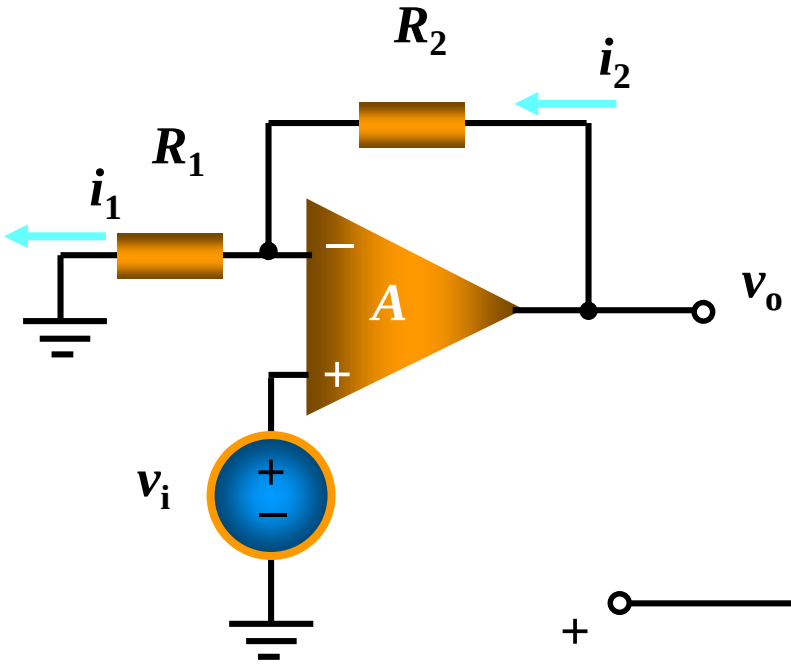
2. 运放的反相和同相接法

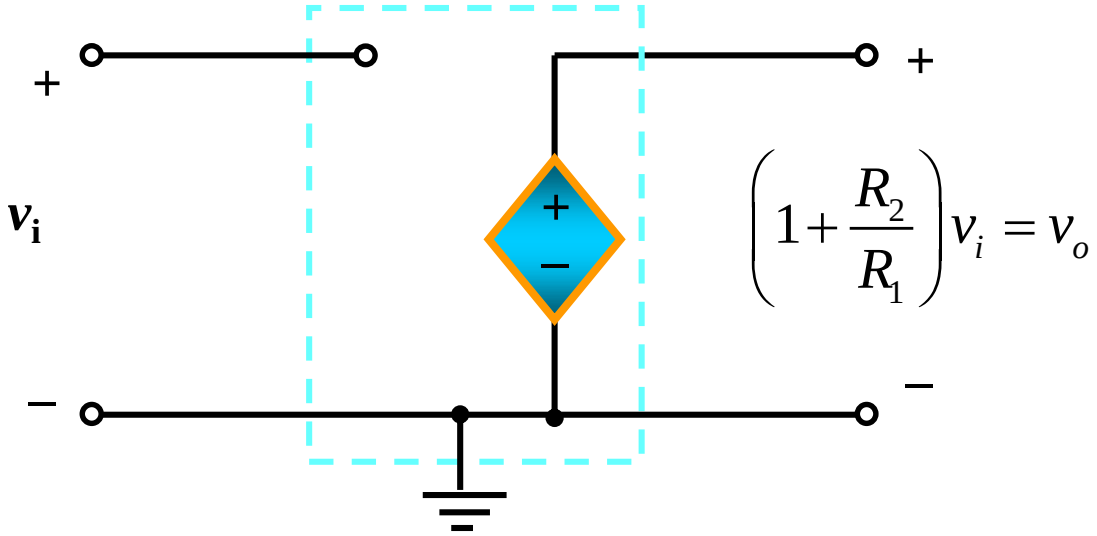
运放的倒相接法电路





运放的同相接法电路





3. 运放典型应用电路

① 加法电路

② 差分放大电路

③ 典型的仪表放大电路

④ 积分电路

⑤ 微分电路

⑥ 模拟电压表电路

⑦ 电压比较器

频率响应

1、一阶无源和有源滤波器

① 低通

② 高通积分电路

③ 带通

④ 带阻

2、增益函数

$$A(s)=A_M F_L(s) F_H(s)$$

$$F_L(s)=\frac{1}{1+\frac{\omega_p}{s}} \qquad F_L(f)=\frac{1}{1+\frac{f_p}{jf}}=\frac{\frac{if}{f_p}}{1+\frac{if}{f_p}}$$

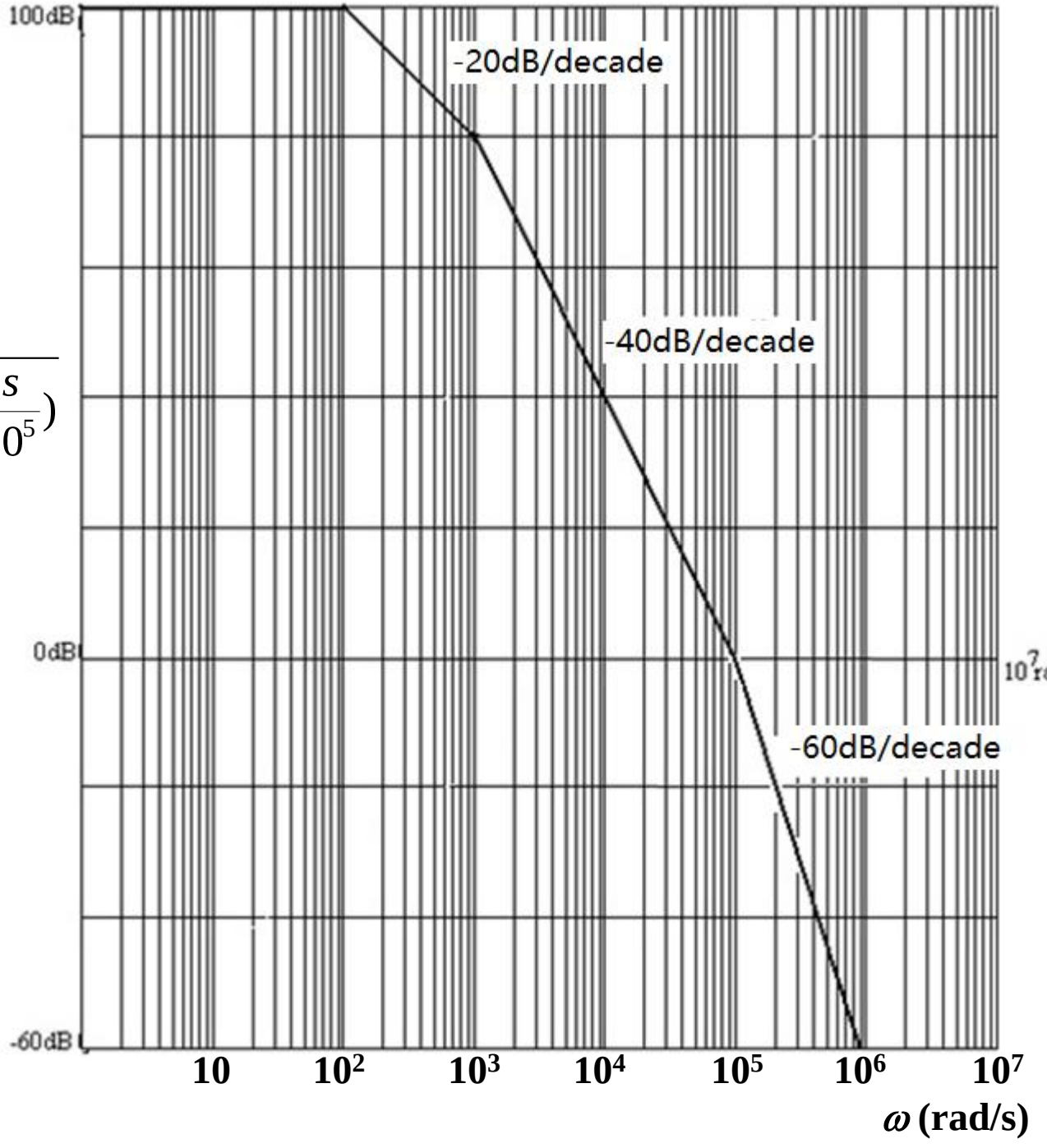
$$F_H(s)=\frac{1}{1+\frac{s}{\omega_p}} \qquad F_H(f)=\frac{1}{1+\frac{jf}{f_p}}$$

3、波特图

例1

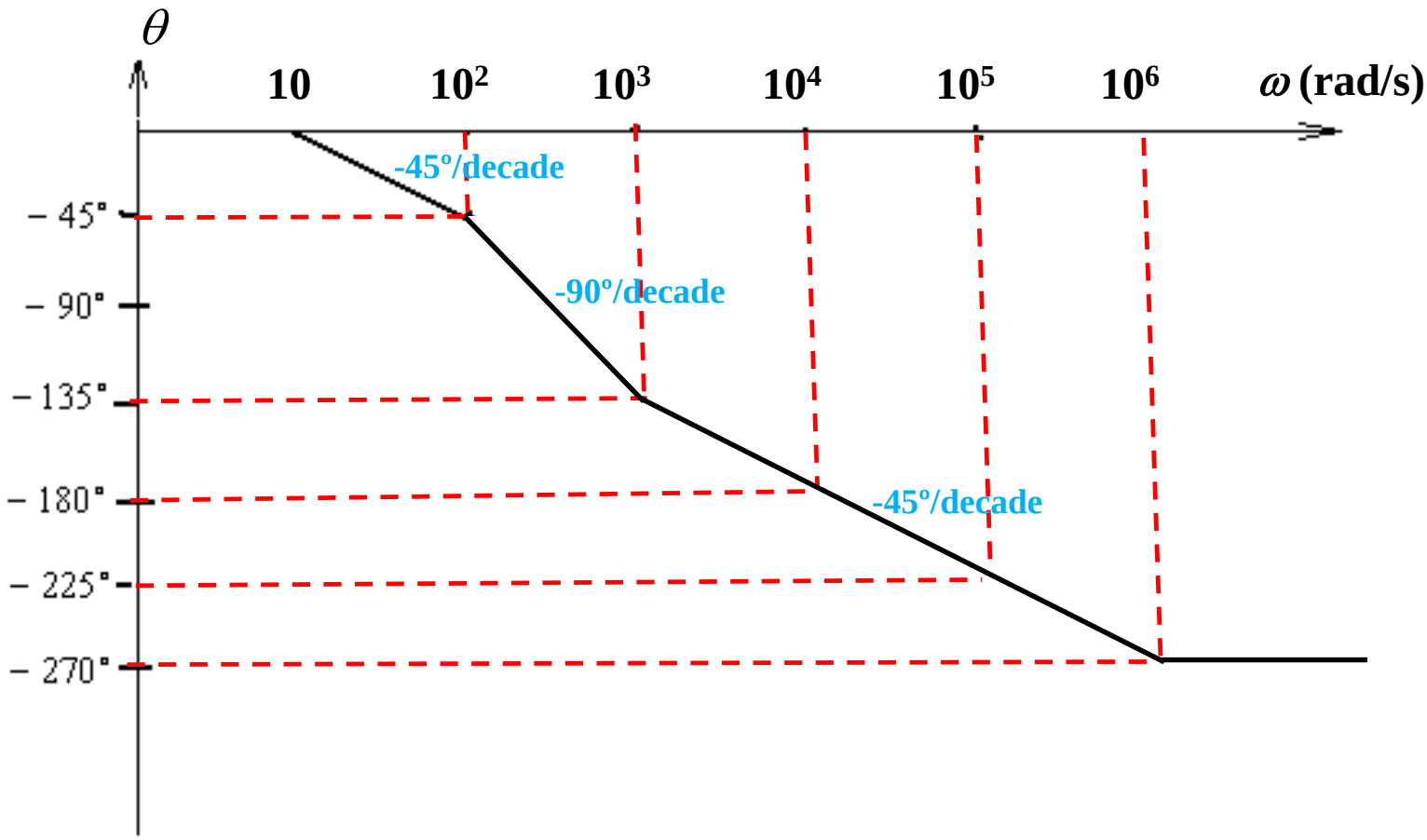
$$T(s)=\frac{10^5}{(1+\frac{s}{10^2})(1+\frac{s}{10^3})(1+\frac{s}{10^5})}$$

幅度波特图



相位波特图

$$T(s)=\frac{10^5}{(1+\frac{s}{10^2})(1+\frac{s}{10^3})(1+\frac{s}{10^5})}$$



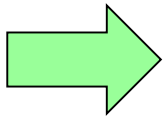
例2 已知某电路电压放大倍数，求它的高、低频截止频率。

$$\dot{A}_v = \frac{-10\,jf}{\left(1+j\frac{f}{10}\right)\left(1+j\frac{f}{10^5}\right)}$$

解

$$\dot{A}_v = \frac{-10\,jf}{\left(1+j\frac{f}{10}\right)\left(1+j\frac{f}{10^5}\right)} = \frac{-100\cdot\frac{jf}{10}}{\left(1+\frac{jf}{10}\right)\left(1+\frac{jf}{10^5}\right)}$$

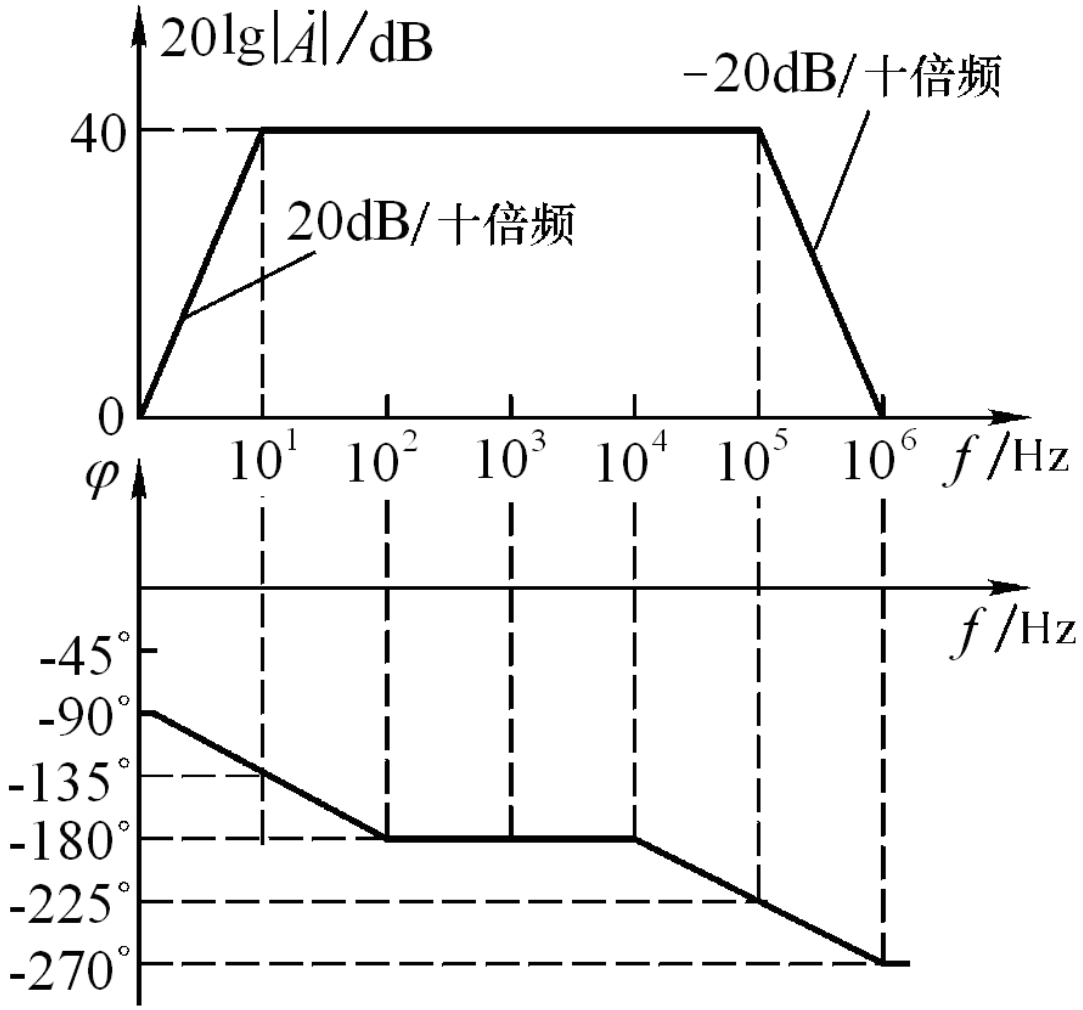
$$= -100\times\frac{1}{1+\frac{10}{jf}}\times\frac{1}{1+\frac{jf}{10^5}}$$



$$f_L = 10\ \text{Hz}$$

$$f_H = 10^5\ \text{Hz}$$

$$\dot{A}_v = -100 \times \frac{1}{1+\frac{10}{jf}} \times \frac{1}{1+\frac{jf}{10^5}}$$



4、电路低频响应特性分析

解题步骤：

1）求解一个电容单独作用时的截止频率，其它电容短路 。

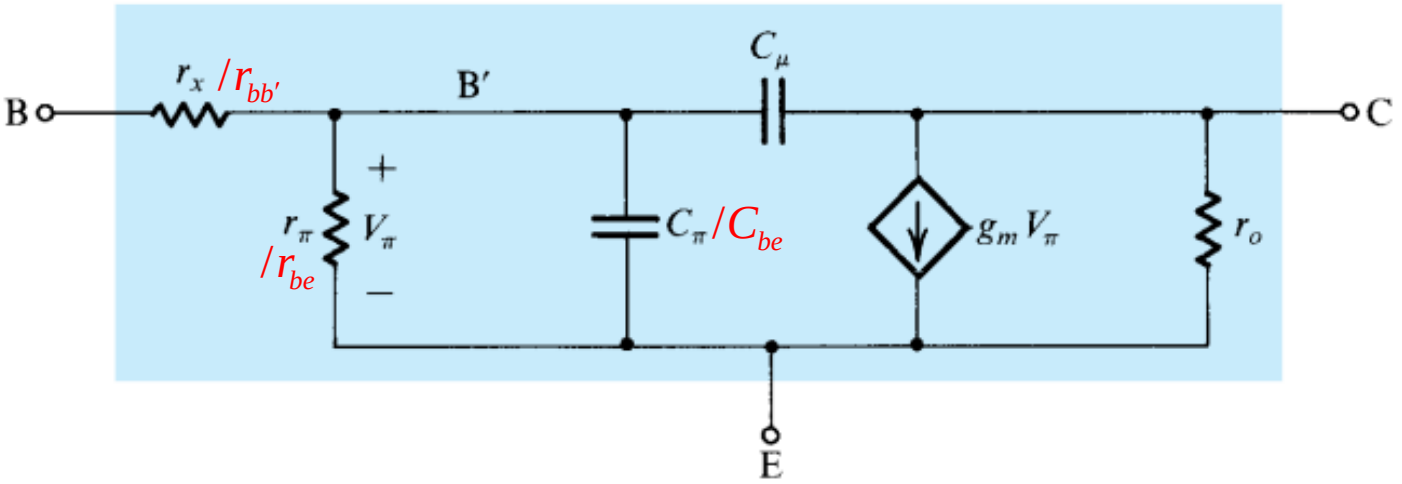
2）确定总的低频截止频率。

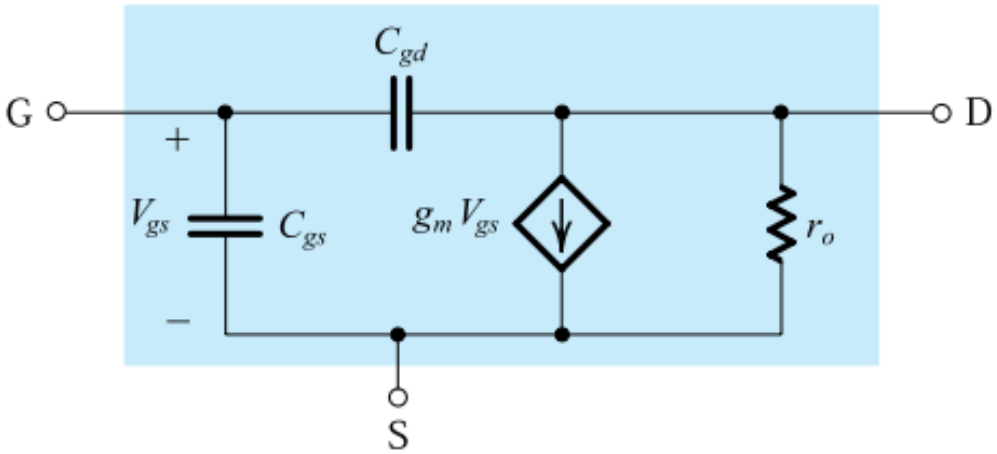
1. 如果每个电容单独作用时计算得到的频率分得很开，其中某个频率占主要地位，则低频截止频率近似为该占主要地位的频率。

2. 若上述条件不满足，则手工计算一般用所有频率的和作为低频截止频率，即

$$f_L = f_{L1} + f_{L2} + f_{L3} + \cdots$$

5、电路高频响应特性分析





解题步骤：

1）求解一个电容单独作用时的截止频率，其它电容开路。

2）确定高频截止频率。

1. 如果每个电容单独作用时计算得到的频率分得很开，其中某个频率占主要地位，则高频截止频率近似为该占主要地位的频率（最小频率）。

2. 若上述条件不满足，则手工计算一般采用以下方法

$$f_H = \frac{1}{\frac{1}{f_{H1}} + \frac{1}{f_{H2}} + \frac{1}{f_{H3}} + \ldots}$$

BJT的内部特性

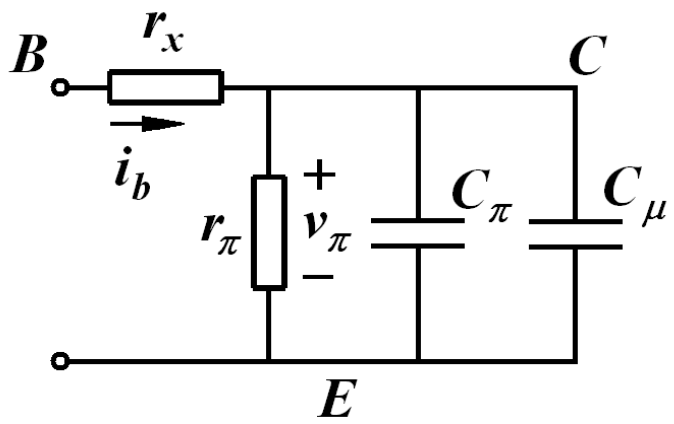


图 7.2.6 (b)

$$\omega_\beta = \frac{1}{r_\pi\left(C_\pi + C_\mu\right)}$$

特征频率

增益带宽积

$$f_T = \beta_0 f_\beta$$

截止频率

米勒效应

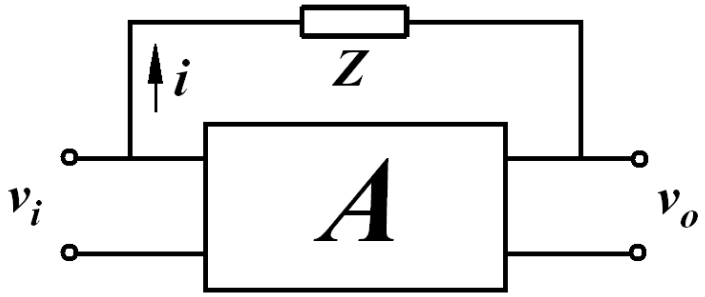


图7.2.8（a）

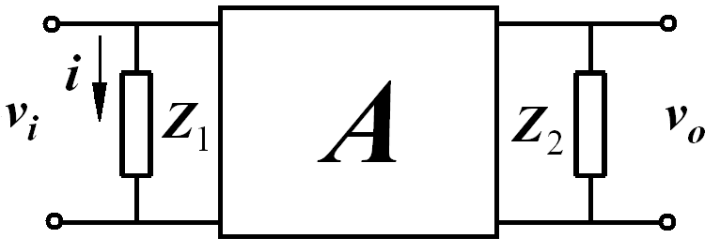
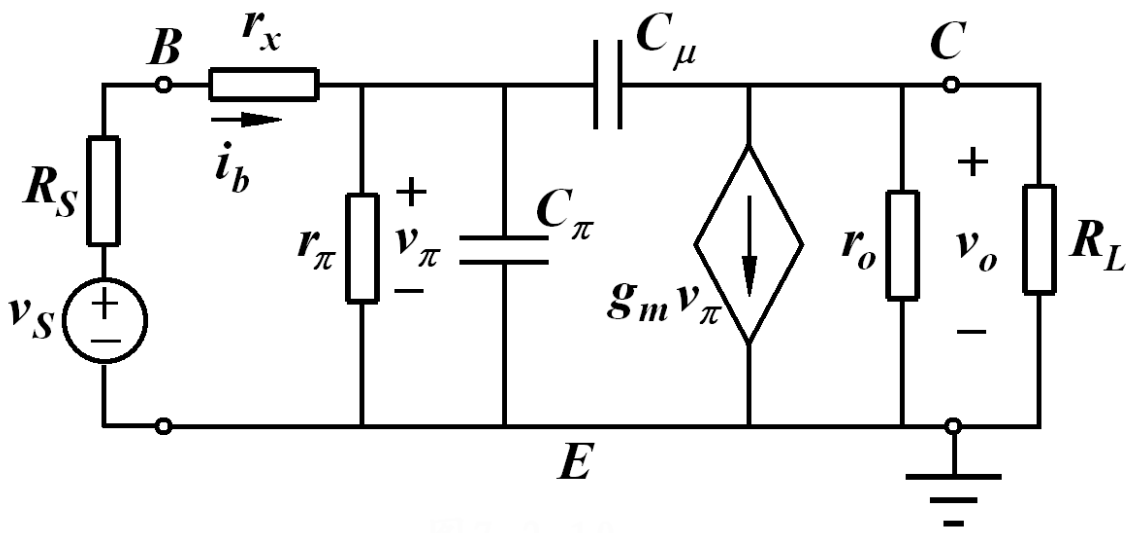


图7.2.8（b）

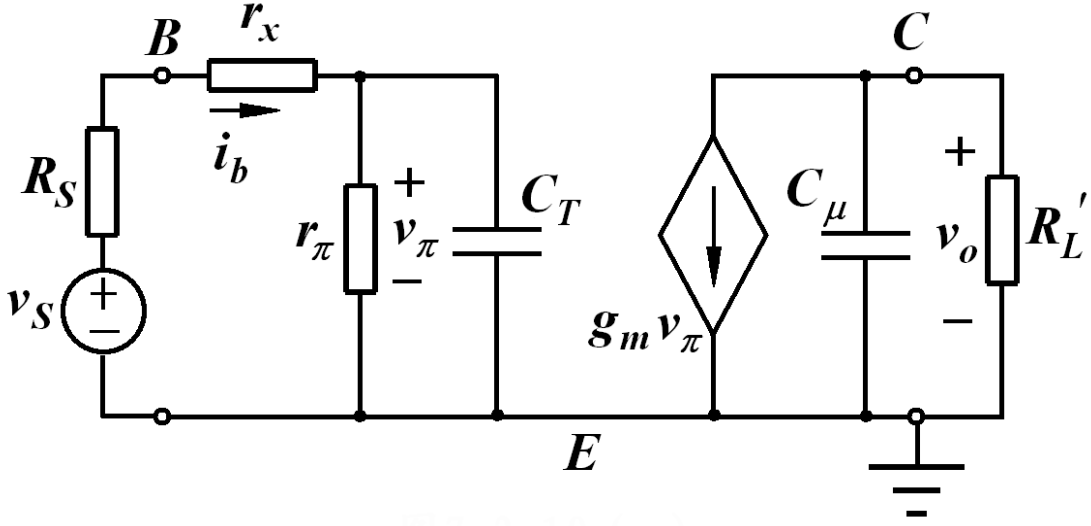
$$Z_1 = \frac{1}{1-A}Z \quad \Rightarrow \quad C_1 = (1-A)C$$

$$Z_2 = \frac{A}{A-1}Z \quad \Rightarrow \quad C_2 = \frac{A}{A-1}C \approx C$$

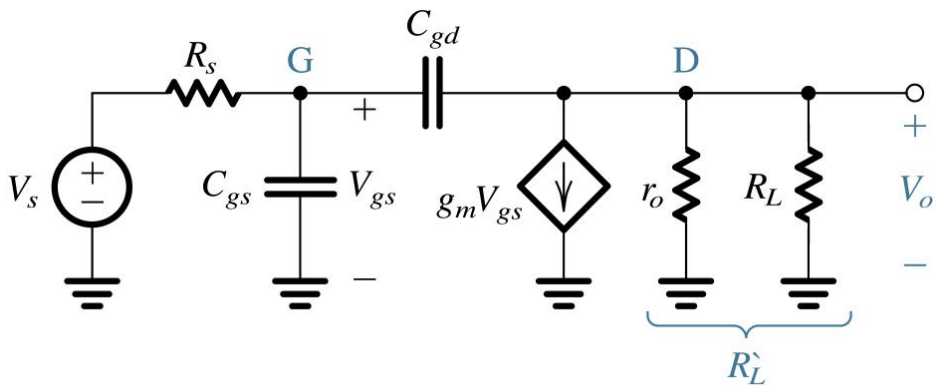
共射放大器的高频响应



$$C_T = C_\pi + C_\mu' = C_\pi + \left(1 + g_m R_L'\right) C_\mu$$

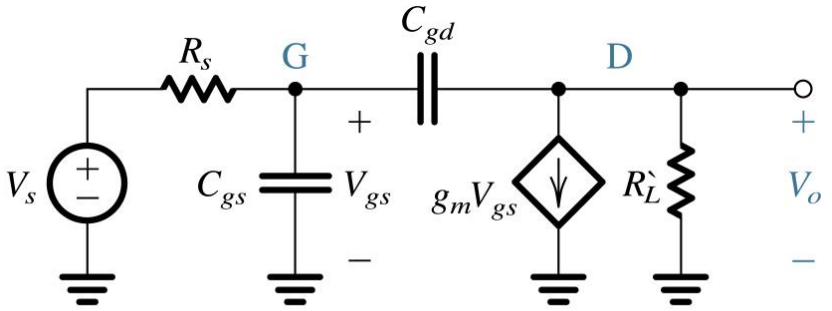


共源放大器的高频响应



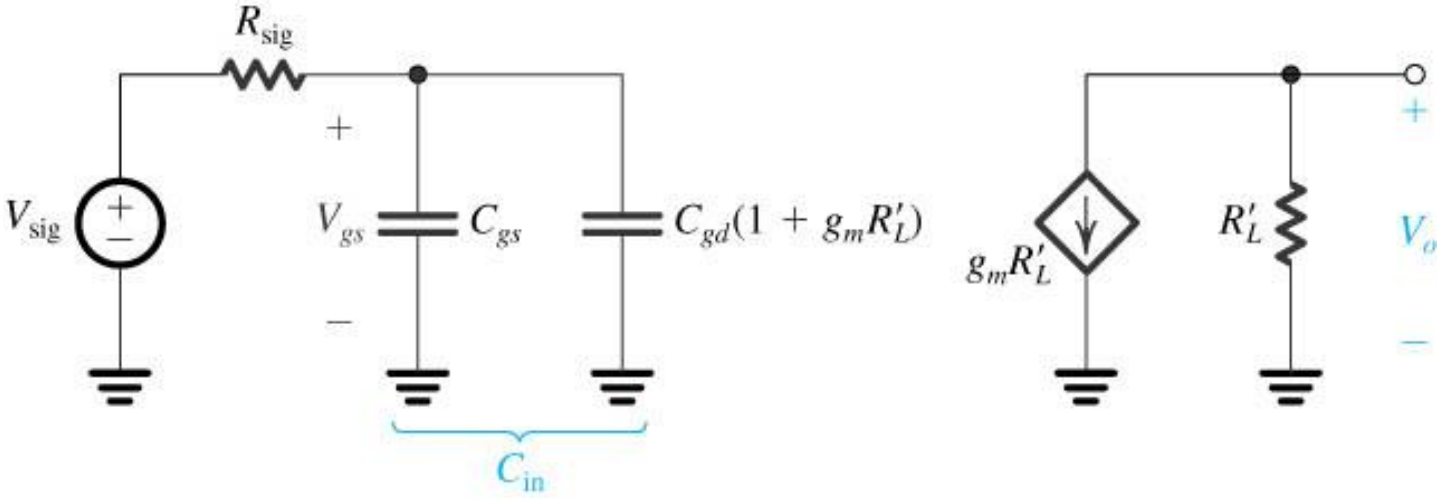
(a)

$$R_L' = r_o // R_L$$



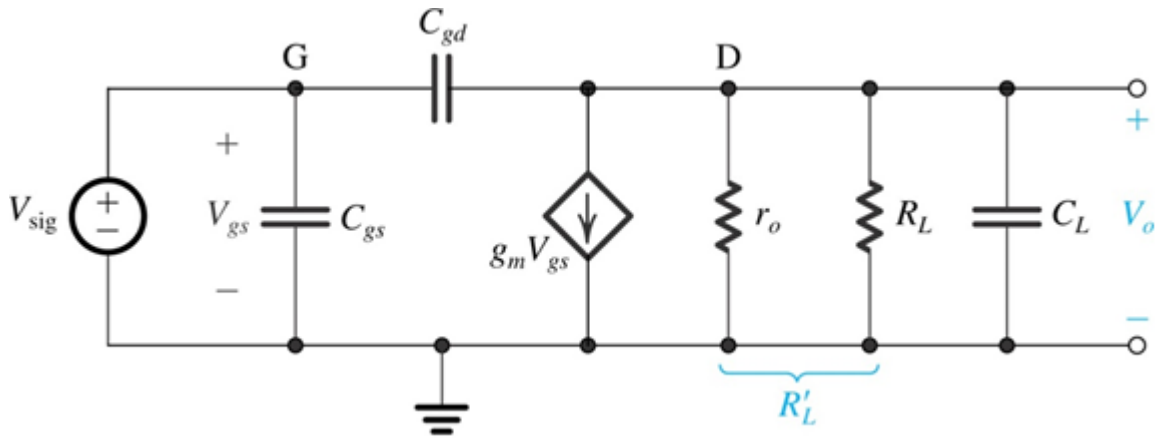
(b)

在R_{sig}很大时



$$C_{eq} = C_{gd}(1 + g_m R_L') \qquad C_T = C_{gs} + C_{gd}(1 + g_m R_L') \qquad f_H = \frac{1}{2\pi C_T R_{sig}}$$

当R_{sig}很小时



$$f_H = \frac{1}{2\pi\left(C_L + C_{gd}\right)R_L'}$$

反馈

1、负反馈对放大器性能的影响

- 负反馈网络对放大器增益的影响

$$A_f = \frac{A}{1+AF} \approx \frac{1}{F}$$（深度负反馈）

- 负反馈网络提高放大器增益稳定性

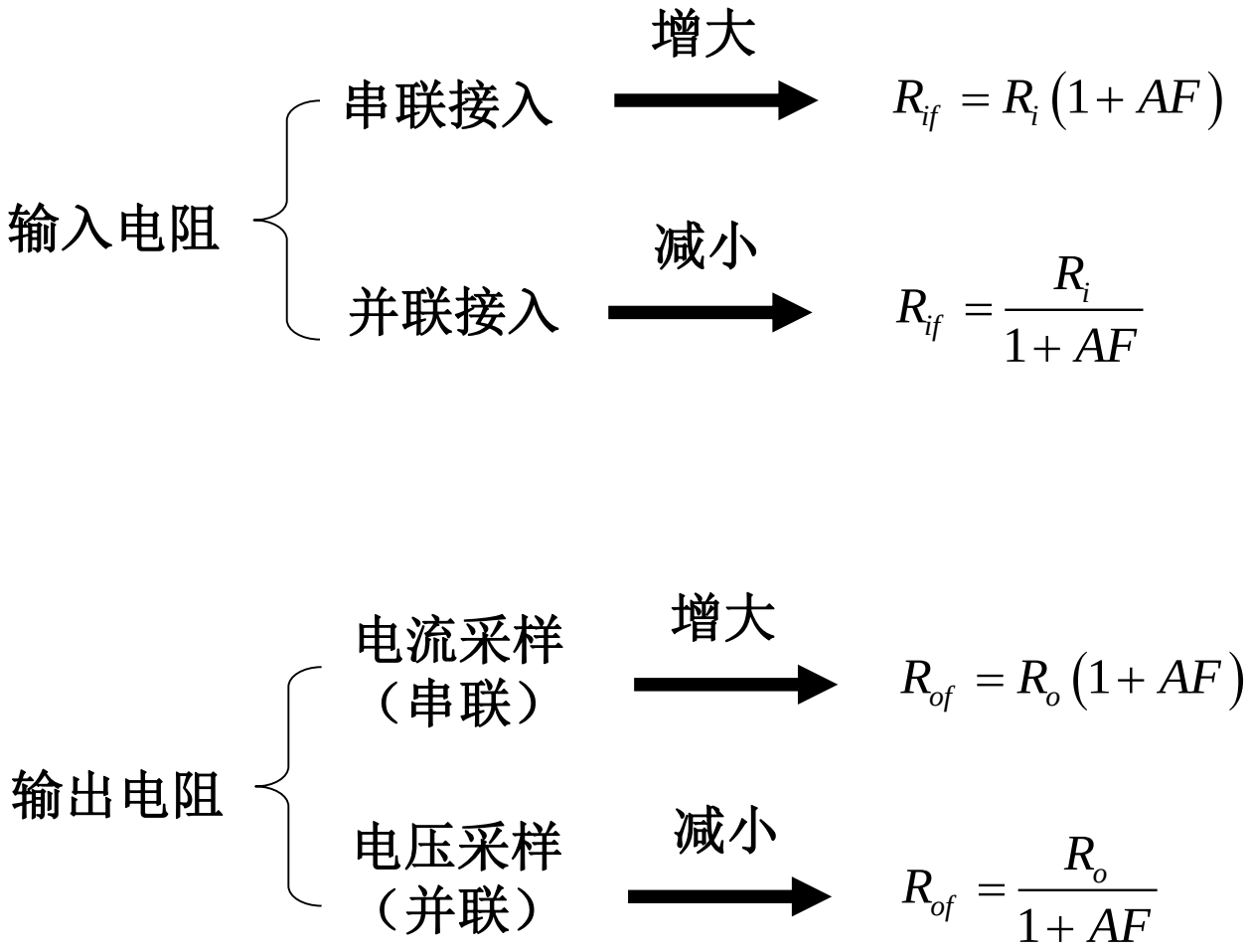
$$\frac{dA_f}{A_f} = \frac{1}{1+AF}\frac{dA}{A}$$

- 负反馈网络扩展放大器的带宽

$$f_{Hf} = (1+A_M F) f_H,\ f_{Lf} = \frac{f_L}{1+A_M F},\ f_{BWf} = (1+A_M F) f_{BW}$$

- 负反馈网络降低放大器的噪声，提高放大器的信噪比

- 负反馈网络减小放大器的非线性失真


输入电阻
串联接入
增大
$R_{if} = R_i\left(1+AF\right)$
并联接入
减小
$R_{if} = \frac{R_i}{1+AF}$
输出电阻
电流采样
（串联）
增大
$R_{of} = R_o\left(1+AF\right)$
电压采样
（并联）
减小
$R_{of} = \frac{R_o}{1+AF}$

2、四种负反馈类型及其判别

① 学会判断，注意输入、输出对地短路点的选择。

② 对开环放大器增加负反馈电路。

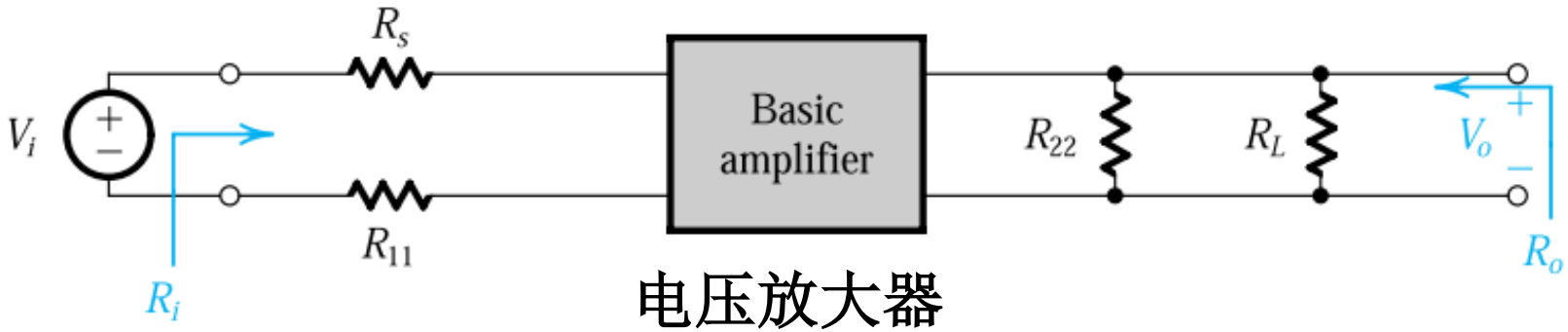
首先，瞬时极性法标注各关键点的极性；

然后，寻找可能的采样点和反馈点；

最后，根据反馈类型确定接入点。

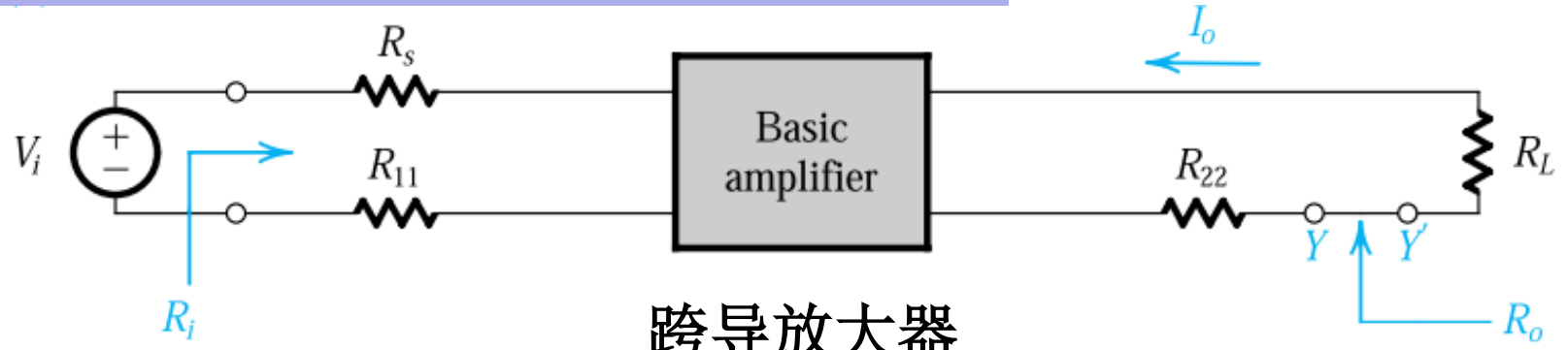
3、负反馈电路分析

电压串联负反馈（串联并联负反馈）



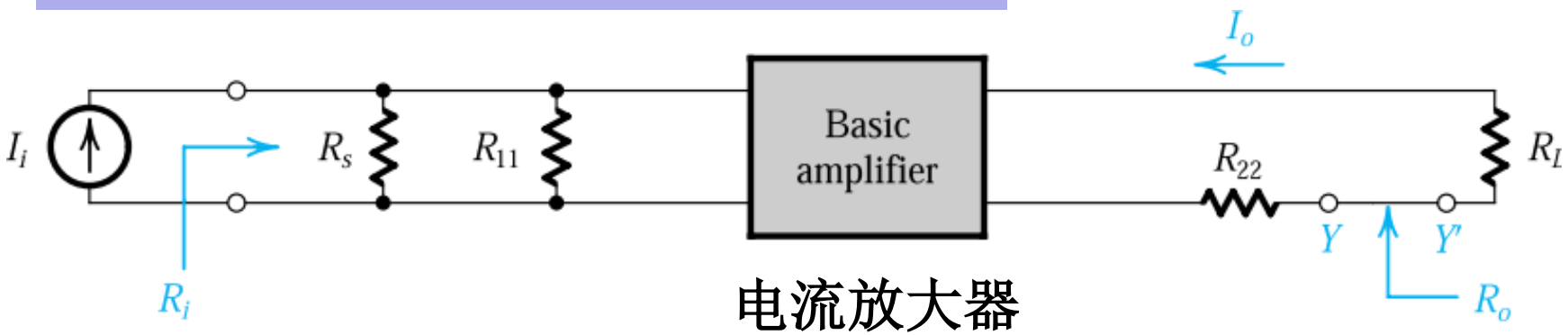
电压放大器

电电流串联负反馈（串联串联负反馈）



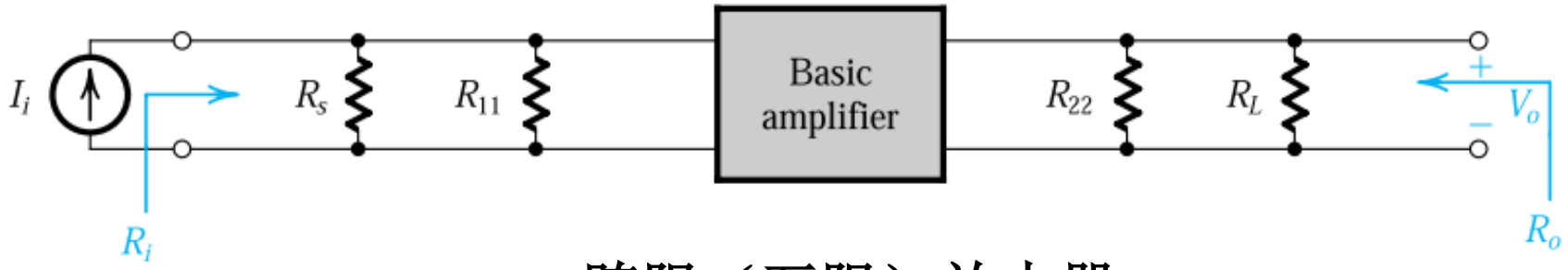
跨导放大器

电流并联负反馈（并联串联负反馈）



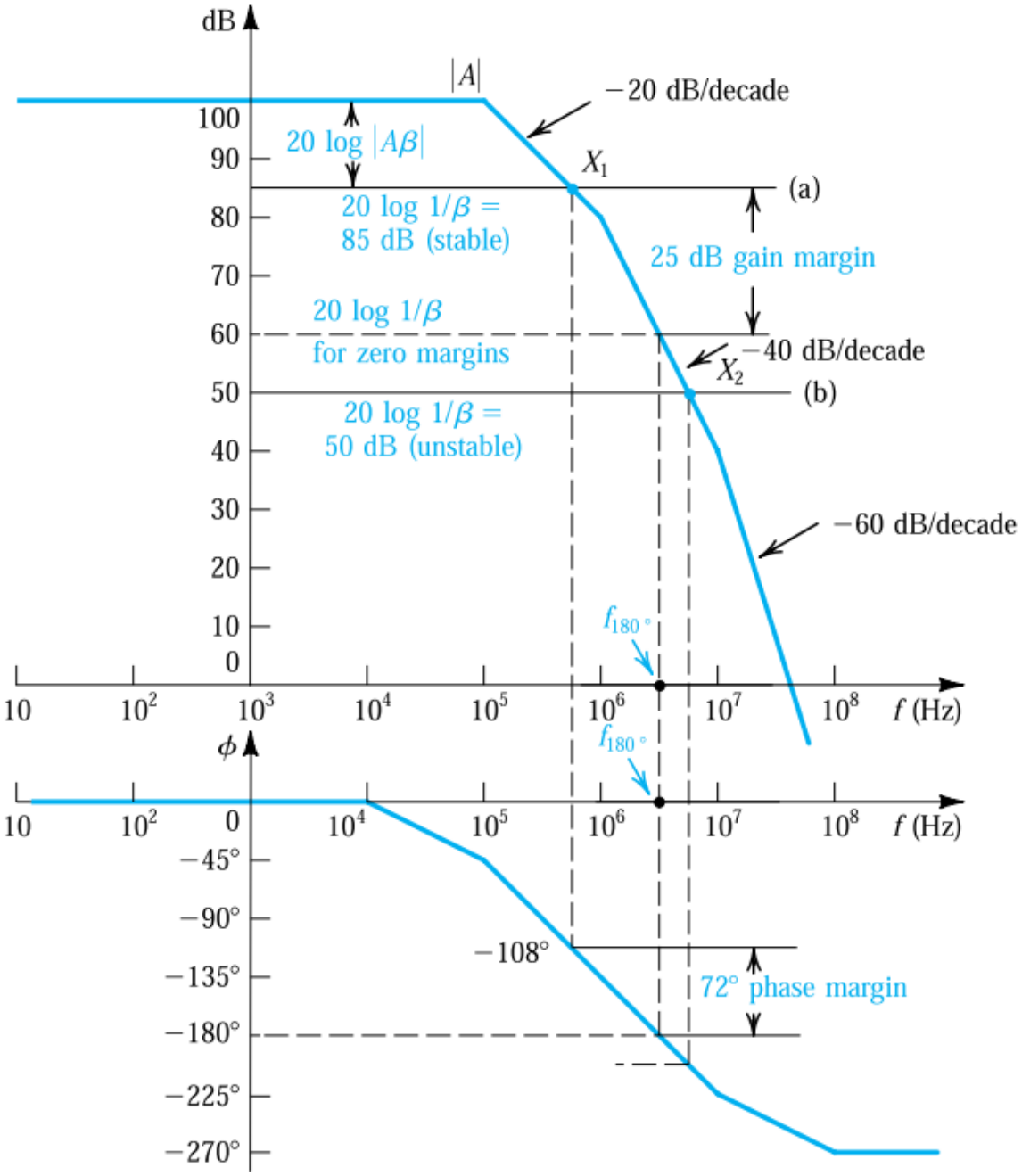
电流放大器

电压并联负反馈（并联并联负反馈）



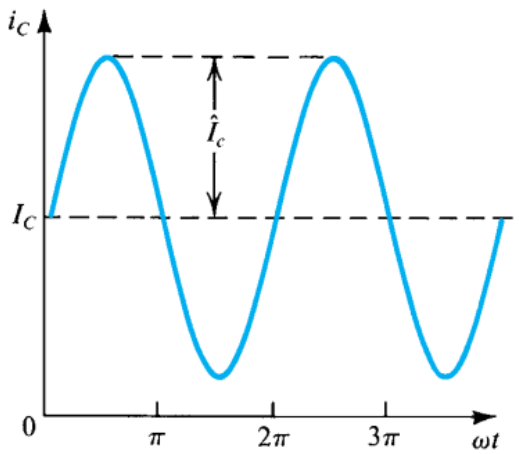
跨阻（互阻）放大器

4、结合波特图分析负反馈电路的稳定性问题

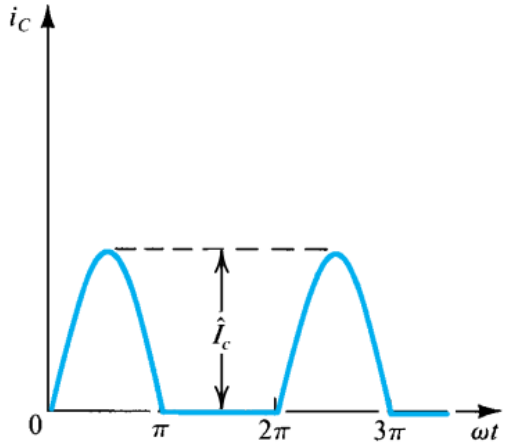


输出级和功率放大器

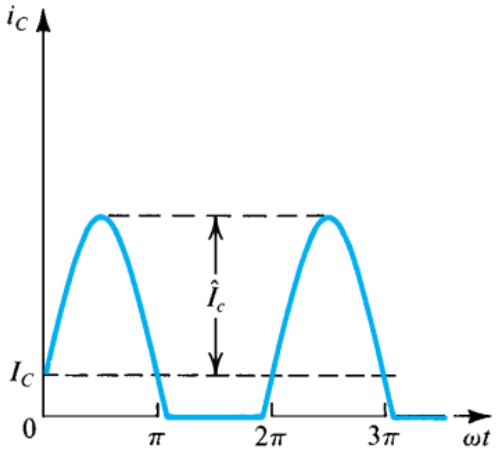
1、放大器的分类

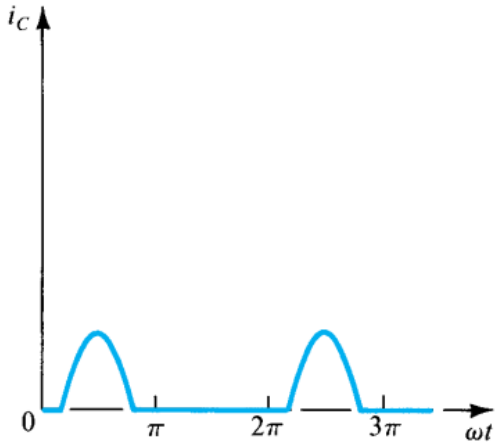


(a)



(b)





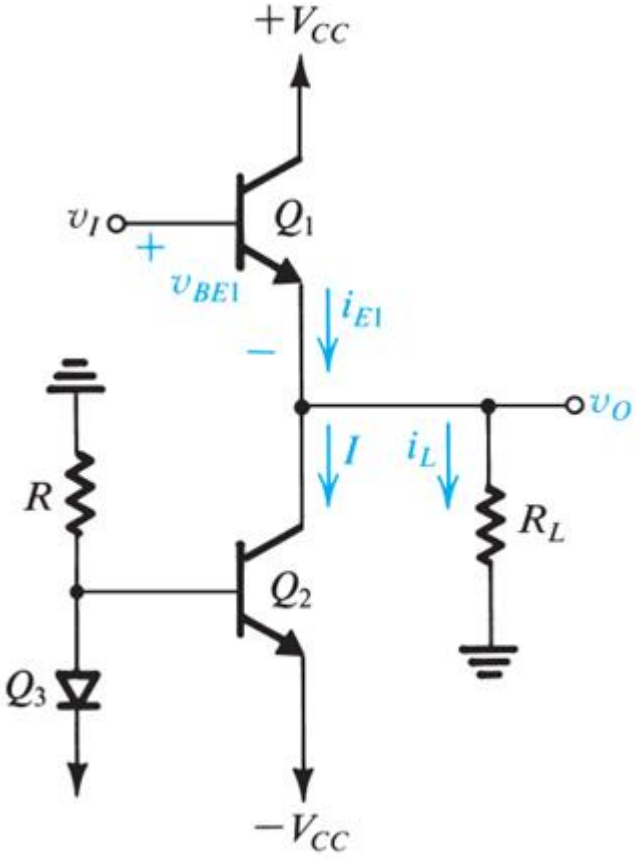
2、放大器电路的分析

1）输入电阻、输出电阻、最大和最小输出电压、输出功率、功率增益、效率

① 注意平均输出功率、电源输出功率、管耗的求解方法。

② 注意晶体管的管压降。

A类输出级电路



因要求 $I \geq \left| \dfrac{-V_{CC} + V_{CE2sat}}{R_L} \right|$ ，忽略饱和压降，则 $I \geq \dfrac{V_{CC}}{R_L}$

若要获得最大功率转换效率，取 $I_{\min} = \dfrac{V_{CC}}{R_L}$

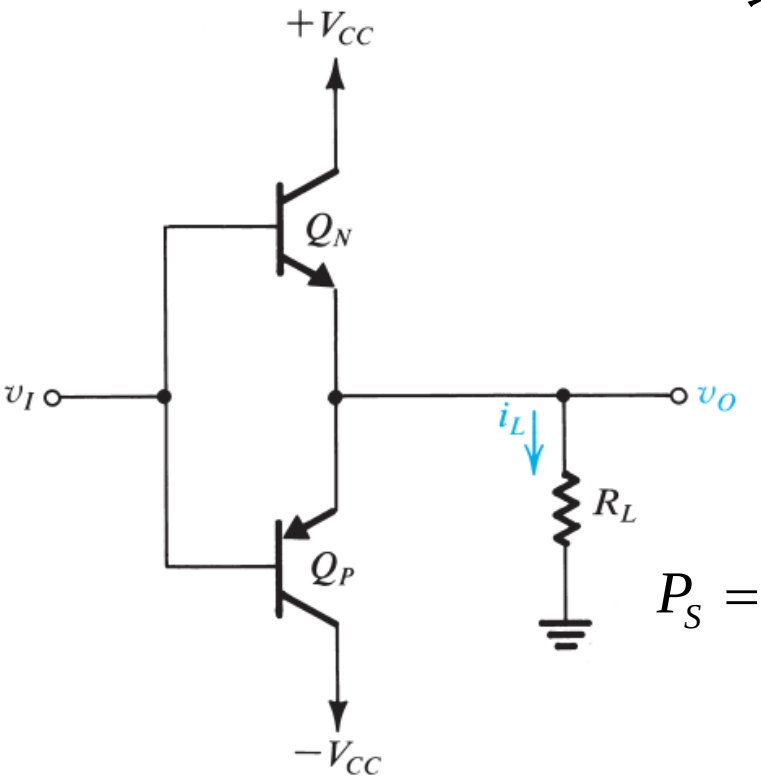
则电源提供的最小平均功率为

$$P_{S\min} = 2V_{CC}I_{\min} = \frac{2V_{CC}^2}{R_L}$$

最大功率转换效率 $\eta_{\max} = \dfrac{P_{L\max}}{P_{S\min}} = 25\%$

此时最大管压降为 $v_{CE} = 2V_{CC}$ ，集电极最大电流为 $i_C = 2I$

B类输出级电路



理想情况下，不考虑饱和压降

$$v_{O\max} = V_{CC} \qquad i_{O\max} = V_{CC}/R_L$$

$$P_L = \frac{1}{2}\frac{\hat{V}_O^{\,2}}{R_L} \xrightarrow{\hat{V}_{O\max} = V_{CC}} P_{L\max}\big|_{\hat{V}_O = V_{CC}} = \frac{1}{2}\frac{V_{CC}^{\,2}}{R_L}$$

$$P_S = P_{S+} + P_{S-} = 2V_{CC}\frac{1}{2\pi}\int_0^{\pi}\frac{\hat{V}_O \sin(\theta)}{R_L}d\theta = \frac{2}{\pi}\frac{V_{CC}}{R_L}\hat{V}_O$$

$$\Downarrow$$

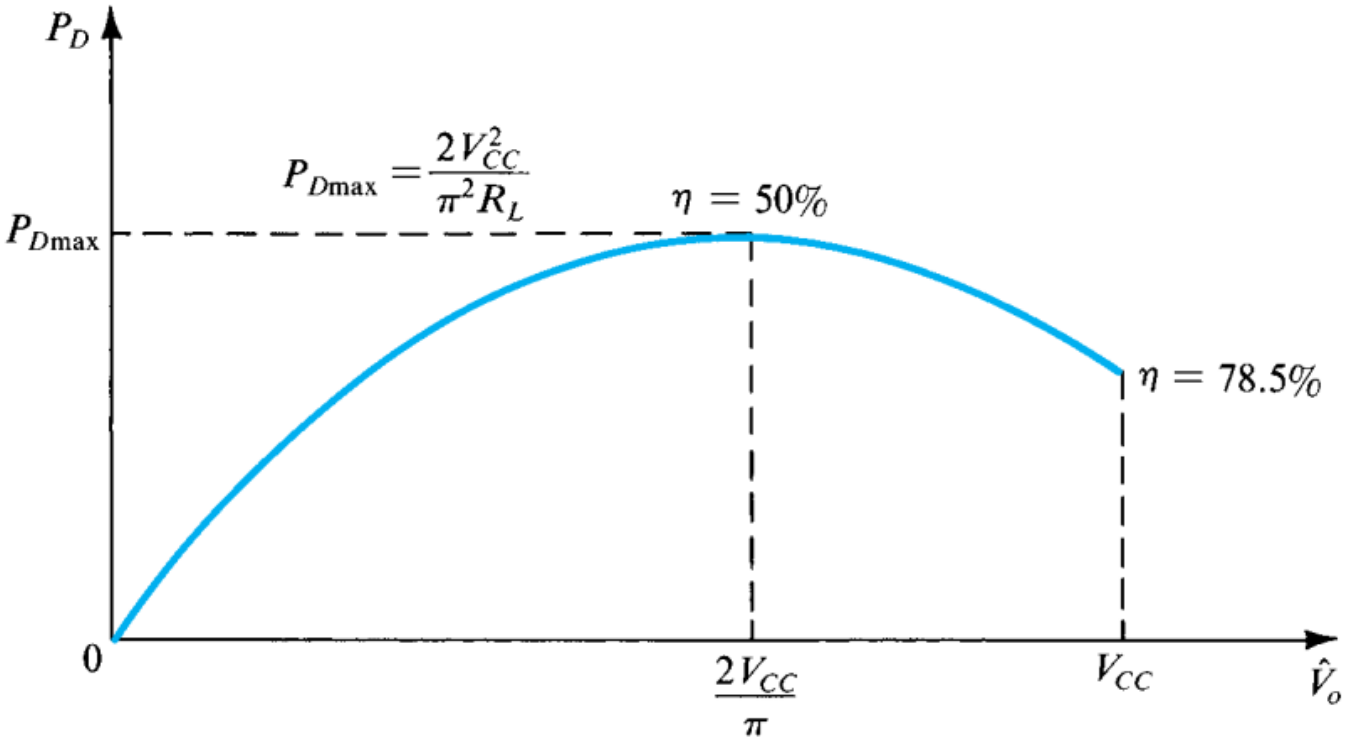
$$P_{S\max}\big|_{\hat{V}_O = V_{CC}} = \frac{2}{\pi}\frac{V_{CC}^{\,2}}{R_L}$$

$$\eta = \frac{P_L}{P_S} = \frac{\pi}{4}\frac{\hat{V}_O}{V_{CC}} \longrightarrow \eta_{\max}\big|_{\hat{V}_O = V_{CC}} = \frac{\pi}{4} \approx 78.5\%$$

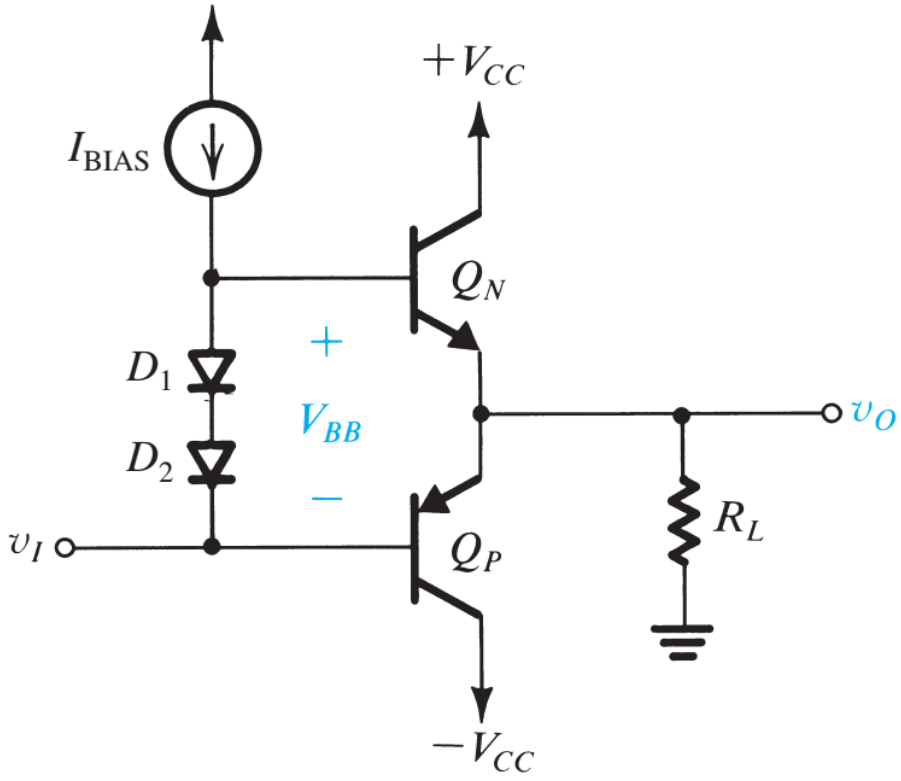
管耗问题

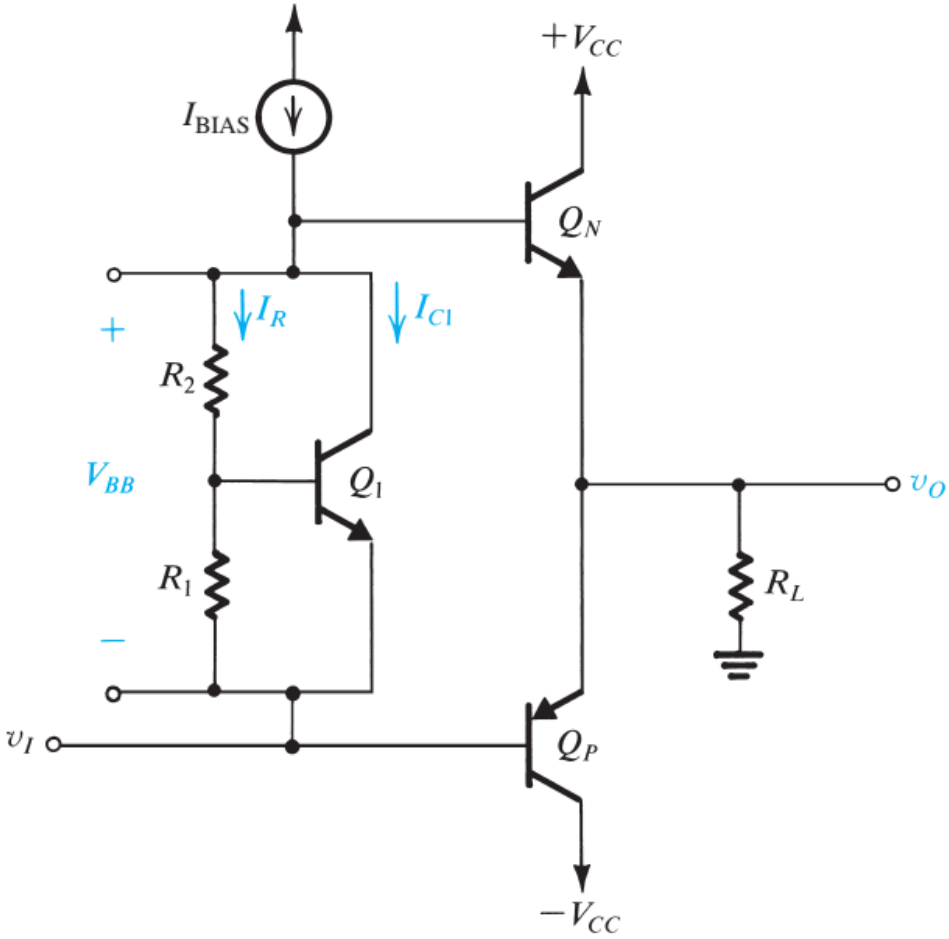
$$P_L = \frac{1}{2}\frac{\hat{V}_O^{\,2}}{R_L} \qquad P_S = \frac{2}{\pi}\frac{\hat{V}_O}{R_L}V_{CC}$$

$$P_D = P_S - P_L = \frac{2}{\pi}\frac{\hat{V}_O}{R_L}V_{CC} - \frac{1}{2}\frac{\hat{V}_O^{\,2}}{R_L}$$



关注AB类放大器消除交越失真的方法





$$V_{BB} = I_R\left(R_1 + R_2\right) = V_{BE}\left(1 + \frac{R_2}{R_1}\right)$$

YΔ电阻关系	ΔY 电阻关系
$R_{12}=\dfrac{R_1R_2+R_2R_3+R_3R_1}{R_3}$ $R_{23}=\dfrac{R_1R_2+R_2R_3+R_3R_1}{R_1}$ $R_{31}=\dfrac{R_1R_2+R_2R_3+R_3R_1}{R_2}$	$R_1=\dfrac{R_{12}R_{31}}{R_{12}+R_{23}+R_{31}}$ $R_2=\dfrac{R_{23}R_{12}}{R_{12}+R_{23}+R_{31}}$ $R_3=\dfrac{R_{31}R_{23}}{R_{12}+R_{23}+R_{31}}$

一阶电路全响应　　$f(t)=f'(t)[f_+(0)-f_+(0)]e^{-\frac{t}{\tau}}$

二阶电路响应：

$R>2\sqrt{\dfrac{L}{C}}$　　$v_C=k_1e^{s_1t}+k_2e^{s_2t}$

$R=2\sqrt{\dfrac{L}{C}}$　　$v_C=k_1e^{-\alpha t}+k_2te^{-\alpha t}$

$R<2\sqrt{\dfrac{L}{C}}$　　$v_C=ke^{-\alpha t}\sin(\omega t+\beta)$

$$i = I_s\left(e^{qv/kT} - 1\right) \qquad i_C = I_S e^{v_{BE}/V_T} \qquad I_S = A_E q D_n n_{p0}/W$$

$$i_D = k_n' \frac{W}{L}\left[\left(v_{GS} - V_t\right)v_{DS} - \frac{1}{2}v_{DS}^2\right] \qquad i_D = \frac{1}{2}k_n' \frac{W}{L}(v_{GS} - V_t)^2$$

$$i_D = \frac{1}{2}k_n' \frac{W}{L}\left(v_{GS} - V_t\right)^2\left(1 + \lambda v_{DS}\right) \qquad \lambda = -\frac{1}{V_A}$$

$$g_m = k_n' \frac{W}{L}\left(V_{GS} - V_t\right) \qquad g_m = \frac{I_D}{\left(V_{GS} - V_t\right)/2}$$